BOUNDEDNESS OF K-SEMISTABLE LOG CONE SINGULARITIES WITH COEFFICIENTS BOUNDED AWAY FROM ZERO

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Abstract. We prove a conjecture of Liu, Moraga, and Süß: in any fixed dimension, K-semistable log Fano cone singularities whose local volumes and boundary coefficients are bounded away from zero are log bounded. The result also holds for boundaries with real coefficients.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

By [Blu18, LX18, Xu20, XZ21, XZ25] for $\mathbb{Q}$-divisors and [Zhu24b, Theorem 2.19] for $\mathbb{R}$-divisors (see also [XZ24, Theorem 2.6]), any klt singularity $(X \ni x, B)$ degenerates, preserving the local volume, to a K-semistable log Fano cone singularity (see Definition 2.5). Thus the local volumes of klt singularities are local volumes of K-semistable log Fano cone singularities. Questions on local volumes, such as [LLX20, Question 6.12], which is formulated for klt singularities $(X \ni x, B)$ with $B \in I$ for a fixed finite set $I \subset [0,1]$ and which [XZ24, Theorem 1.2] answers positively, lead to the question of whether K-semistable log Fano cone singularities of fixed dimension, with local volumes bounded from below, form a bounded family. Liu, Moraga, and Süß proposed the following conjecture, which also assumes a positive lower bound on the coefficients of the boundary.

Conjecture 1.1 ([LMS26, Conjecture 1])**.** *Let d be a positive integer and ϵ, η two positive real numbers. Then the class of d-dimensional K-semistable singularities $(X \ni x, B)$ such that $\widehat{\mathrm{vol}}(X \ni x, B) > \epsilon$ and $\mathrm{mult}_P B > \eta$ for any prime divisor $P \subset \mathrm{Supp}\, B$ is log bounded.*

In Conjecture 1.1 we follow the terminology of [LMS26]. Let X be an affine variety with an effective action of an algebraic torus T such that the weights of the T-action on the coordinate ring of X span a strictly convex cone of dimension $\dim T$, and the only T-invariant regular functions on X are the constants. Then T has a unique fixed point $x \in X$. If moreover B is a T-invariant $\mathbb{Q}$-divisor on X such that $(X \ni x, B)$ is klt, then $(X \ni x, B)$ is called a *log Fano cone singularity* in [LMS26, Definition 2.7].

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A K-semistable singularity is a log Fano cone singularity in the sense of [LMS26, Definition 2.7] that is K-semistable with respect to some polarization [LMS26, Definition 2.16]. In [LMS26, Definition 2.17], log bounded classes of pairs are defined first, and a class of K-semistable singularities is called *log bounded* if there exists a log bounded class of pairs such that for any member $(X \ni x, B)$ of the class of singularities, the germ of (X, B) at x is isomorphic to the germ of some pair in the class of pairs at some point. We refer the reader to Definition-Lemma 2.2 for local volumes.

See also [XZ25, Conjecture 1.7] and [Zhu24b, Conjecture 1.1], which assume that $B \in I$ for a fixed finite set $I \subset [0,1]$ ($I \subset [0,1] \cap \mathbb{Q}$ in [XZ25]) instead of a lower bound on the coefficients of B. A related conjecture considers all klt singularities: Han, Liu, and Qi [HLQ23, Conjecture 8.9] conjectured that the d-dimensional klt singularities $(X \ni x, B)$ such that $\widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon$ and $\mathrm{mult}_P B \geq \eta$ for any prime divisor $P \subset \operatorname{Supp} B$ are log bounded up to special degeneration in the sense of [HLQ23, Definition 8.7].

Conjecture 1.1 was proved in the following cases.

(1) [XZ24, Theorem 1.1] proved Conjecture 1.1 when $B \in I$ for a fixed finite set $I \subset [0,1]$ (the postscript in [LMS26, p. 3036] states that Conjecture 1.1 was solved in [XZ24]; [XZ24, Theorem 1.1] assumes that $B \in I$ for a finite set I). Before [XZ24], this case was proved when $d = 3$ by [LMS26, Theorem 1] and [Zhu24b, Corollary 1.4], and for singularities of complexity one, that is, with an effective action of a torus of dimension $d-1$ [LMS26, Definition 2.20], by [LMS26, Theorem 3].
(2) [LMS26, Theorem 6] proved Conjecture 1.1 when $B = 0$ and $X \ni x$ is a hypersurface singularity.
(3) Conjecture 1.1 holds when $B = 0$ and $X \ni x$ is a toric singularity, since then K_X is $\mathbb{Q}$-Cartier and there exist only finitely many d-dimensional $\mathbb{Q}$-Gorenstein toric singularities $X \ni x$ with $\widehat{\mathrm{vol}}(X \ni x, 0) > \epsilon$ [MS24, Theorem 1].

Liu, Moraga, and Süß also note that Conjecture 1.1 is known for exceptional singularities [LMS26, p. 3034], referring to [HLS26].

We now explain why the existing methods do not prove Conjecture 1.1. [XZ24, Proposition 3.3] assumes $B \in I$ for a finite set $I \subset [0,1] \cap \mathbb{Q}$ and uses a positive integer m, depending on I and the other data, with $mI \subset \mathbb{N}$. In Conjecture 1.1, the coefficients of the boundaries in the class may take infinitely many values with unbounded denominators, so no such m exists.

A known way to replace a lower bound on the coefficients by finitely many coefficients is to enlarge the boundary to a boundary with bounded denominators. For any klt germ $(X \ni x, B)$ of dimension d such that $\widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon$ and $\mathrm{mult}_P B \geq \eta$ for any prime divisor $P \subset \operatorname{Supp} B$, [HLQ24, Proposition 5.1] constructs a $\mathbb{Q}$-divisor $B^+ \geq (1+\delta)B$ such that NB^+ is a Weil divisor and $\widehat{\mathrm{vol}}(X \ni x, B^+) \geq \epsilon_0$, where the positive integer N and the positive real numbers δ and ϵ_0 depend only on d, ϵ, and η. [Zhu24a, Remark 4.9] uses such an enlargement to replace the assumption $B \in I$, with I finite, in [Zhu24a, Conjecture 1.7] by a positive lower bound on the coefficients of B.

For log Fano cone singularities, such an enlargement does not directly lead to Conjecture 1.1. For a finite set $I \subset [0,1]$, [XZ24, Theorem 1.4] shows that the d-dimensional polarized log Fano cone singularities $(X \ni x, B; \xi)$ with $B \in I$ whose local volumes and volume ratios (see Definition 2.5) are bounded from below by fixed positive numbers form a bounded set. To use [XZ24, Theorem 1.4] for a log Fano cone singularity $(X \ni x, B)$, the enlarged boundary has to be invariant under the torus T acting on X (see Definition 2.5), while B^+ in [HLQ24, Proposition 5.1] is obtained by adding a general member of a linear system and need not be T-invariant. A T-invariant way to obtain bounded denominators is to write $B = \sum_i b_i D_i$, where the D_i are distinct prime divisors, and to set

$$\bar{B} := \sum_i \frac{1}{q} \lfloor q(1+\tau) b_i \rfloor D_i$$

for a fixed positive integer q and a fixed positive real number τ. Then $\bar{B}$ is T-invariant and $q\bar{B}$ is a Weil divisor, but $K_X + \bar{B}$ need not be $\mathbb{Q}$-Cartier even if $K_X + B$ is. For example, let $X = \{z_1z_2 = z_3z_4\} \subset \mathbb{A}^4$, which is a toric variety, and let $D_1 = \{z_1 = z_3 = 0\}$, $D_2 = \{z_1 = z_4 = 0\}$, $D_3 = \{z_2 = z_3 = 0\}$, and $D_4 = \{z_2 = z_4 = 0\}$ be the torus-invariant prime divisors on X. Since X is a hypersurface in $\mathbb{A}^4$, $K_X \sim 0$. The class group of the local ring $\mathcal{O}_{X,0}$ of X at the origin 0 is $\mathbb{Z}$, generated by $[D_1] = -[D_2] = -[D_3] = [D_4]$ [Har77, II, Exercises 6.3 and 6.5]. Thus, for rational numbers b_1, b_2, b_3, b_4, the divisor $K_X + \sum_i b_iD_i$ is $\mathbb{Q}$-Cartier if and only if $b_1 + b_4 = b_2 + b_3$. Let $(b_1, b_2, b_3, b_4) = (\frac{1}{2}, \frac{2}{5}, \frac{3}{10}, \frac{1}{5})$, $q = 20$, and $\tau = \frac{1}{10}$. Then $b_1 + b_4 = b_2 + b_3$, while

$$\bar{B} = \frac{11}{20}D_1 + \frac{8}{20}D_2 + \frac{6}{20}D_3 + \frac{4}{20}D_4 \geq B$$

and $\frac{11}{20} + \frac{4}{20} \neq \frac{8}{20} + \frac{6}{20}$.

The goal of this paper is to prove Conjecture 1.1 in full generality, that is, with only a positive lower bound on the coefficients of B (see Theorem 1.2).

Theorem 1.2. *Let d be a positive integer and ϵ, η two positive real numbers. Then the set of triples $(X, \operatorname{Supp} B; x)$, where $(X \ni x, B)$ is a d-dimensional K-semistable log Fano cone singularity (see Definition 2.5) such that $\widehat{\operatorname{vol}}(X \ni x, B) \geq \epsilon$ and $\operatorname{mult}_P B \geq \eta$ for any prime divisor $P \subset \operatorname{Supp} B$, is log bounded (see Definition 2.7).*

We obtain the following corollary.

Corollary 1.3. *Conjecture 1.1 holds.*

A K-semistable singularity in the sense of [LMS26, Definition 2.16] is a K-semistable log Fano cone singularity as in Definition 2.5, and log boundedness in the sense of Definition 2.7 implies log boundedness in the sense of [LMS26, Definition 2.17] (see Section 6). Conjecture 1.1 considers $\mathbb{Q}$-divisors B (see [LMS26, Definition 2.7]), while B is an $\mathbb{R}$-divisor in Theorem 1.2. Moreover, the conclusion of Theorem 1.2 is the boundedness of the affine varieties X together with $\operatorname{Supp} B$ and the vertex x, not only of the germs $X \ni x$.

The lower bound η in Theorem 1.2 cannot be removed. Let $m \geq 2$ be an integer, z_1, z_2 coordinates on $\mathbb{A}^2$, $L_1, \dots, L_m \subset \mathbb{A}^2$ distinct lines through the origin 0, ℓ_i a linear equation of L_i for each i, and $B_m := \frac{1}{m}\sum_{i=1}^m L_i$. Let $\mathbb{G}_m$ act on $\mathbb{A}^2$ by scalar multiplication, ξ the Reeb vector on $\mathbb{A}^2$ with $\langle \xi, 1\rangle = 1$, and E the exceptional divisor of the blow-up of $\mathbb{A}^2$ at 0. Then wt_ξ is the valuation given by E, and

$$a(E, \mathbb{A}^2, B_m) = 2 - 1 = 1 = \frac{1}{2}a(E, \mathbb{A}^2, 0).$$

The blow-up of $\mathbb{A}^2$ at 0 is a log resolution of $(\mathbb{A}^2, B_m)$ on which the strict transforms of the L_i are disjoint. Since $a(E, \mathbb{A}^2, B_m) > 0$ and $\frac{1}{m} < 1$, $(\mathbb{A}^2, B_m)$ is klt [KM98, Corollary 2.32(2)]. Since the $\mathbb{G}_m$-action on $\mathbb{A}^2$ is good with vertex 0 and B_m is $\mathbb{G}_m$-invariant, $(\mathbb{A}^2 \ni 0, B_m; \xi)$ is a polarized log Fano cone singularity. Let v be a real valuation centered at 0 and $\mu := \min\{v(z_1), v(z_2)\}$. Then $v(\ell_i) \geq \mu$ for any i. Since any two distinct ℓ_i, ℓ_j span the linear forms in z_1, z_2, we have $v(\ell_i) > \mu$ for at most one i. Let i_0 be an index such that $v(\ell_{i_0})$ is maximal, and let j be an index with $j \neq i_0$, so that $v(\ell_j) = \mu$. Since $(\mathbb{A}^2, L_{i_0} + L_j)$ is lc [KM98, Corollary 2.31(3)], Lemma 2.3(1) and (3) imply

$$a(v, \mathbb{A}^2, 0) \geq v(\ell_{i_0}) + v(\ell_j) = 2\mu + (v(\ell_{i_0}) - \mu).$$

Since $v(B_m) = \mu + \frac{1}{m}(v(\ell_{i_0}) - \mu)$ and $m \geq 2$, we have $v(B_m) \leq \frac{1}{2}a(v, \mathbb{A}^2, 0)$, that is, by Lemma 2.3(1),

$$a(v, \mathbb{A}^2, B_m) \geq \frac{1}{2}a(v, \mathbb{A}^2, 0).$$

As $\widehat{\operatorname{vol}}_{\mathbb{A}^2, B}(v) = a(v, \mathbb{A}^2, B)^2 \operatorname{vol}(v)$ for $B = B_m$ and $B = 0$ when $a(v, \mathbb{A}^2, 0) < +\infty$, and both normalized volumes are $+\infty$ otherwise (Lemma 2.3(1)), we get

$$\widehat{\operatorname{vol}}_{\mathbb{A}^2, B_m}(v) \geq \frac{1}{4}\widehat{\operatorname{vol}}_{\mathbb{A}^2, 0}(v).$$

By [Li18, Proposition 5.2], $\widehat{\mathrm{vol}}_{\mathbb{A}^2,0}(v) \geq 4$, so $\widehat{\mathrm{vol}}_{\mathbb{A}^2,B_m}(v) \geq 1$, with equality when $v = \mathrm{wt}_\xi$. Thus

$$\widehat{\mathrm{vol}}(\mathbb{A}^2 \ni 0, B_m) = 1 = \widehat{\mathrm{vol}}_{\mathbb{A}^2,B_m}(\mathrm{wt}_\xi),$$

and $(\mathbb{A}^2 \ni 0, B_m; \xi)$ is K-semistable by Definition 2.5. However, $\operatorname{Supp} B_m$ has m irreducible components, so the triples $(\mathbb{A}^2, \operatorname{Supp} B_m; 0)$ do not form a log bounded set (in a family as in Definition 2.7, the number of irreducible components of $\mathcal{D} \cap \mathcal{X}_t$, $t \in S$, is bounded, since it is a locally constructible function of t [Gro66, Corollaire 9.7.9]).

Sketch of the proof. Fix $(X \ni x, B)$ in Theorem 1.2 and a Reeb vector ξ with acting torus T such that $(X \ni x, B; \xi)$ is K-semistable. By Proposition 5.1, which uses [XZ24, Theorem 1.4] and Lemma 5.3, it suffices to construct a T-invariant $\mathbb{Q}$-divisor $B^+ \geq B$ such that NB^+ is a Weil divisor, (X, B^+) is klt, and $\widehat{\mathrm{vol}}(X \ni x, B^+) \geq 2^{-d}\epsilon$, where N depends only on d, ϵ, η. We construct B^+ in two steps. (1) Proposition 3.1 with $\lambda = 2/c_0$, for the constant c_0 of Lemma 4.1, provides N and $s \in H^0(X, \mathcal{O}_X)$ such that $G = \frac{1}{N}\operatorname{div}(s) - (K_X + B) \geq 0$ and $(X, B + \lambda G)$ is klt near x. As λ depends only on d, N depends only on d, ϵ, η. (2) Proposition 4.3 sets $B^+ = \frac{1}{N}\operatorname{div}(s_\alpha) - K_X$ for a component s_α of s of minimal ξ-weight. Since $\mathrm{wt}_\xi(s_\alpha) = \mathrm{wt}_\xi(s)$, we have

$$\lambda\,\mathrm{wt}_\xi(B^+ - B) = \lambda\,\mathrm{wt}_\xi(G) \leq a(\mathrm{wt}_\xi, X, B),$$

and Lemma 4.1, applied at wt_ξ, a minimizer of $\widehat{\mathrm{vol}}_{X,B}$, turns it into $a(\cdot, X, B^+) \geq \frac{1}{2}a(\cdot, X, B)$. Step (2) is new, and step (1) adapts [HLQ24, Proof of Proposition 5.1] to any λ and to sections on X (Remark 3.2).

(a) Torus-invariant boundaries. As explained after Conjecture 1.1, the rounding $\bar{B}$ of B is T-invariant but $K_X + \bar{B}$ need not be $\mathbb{Q}$-Cartier, and the boundary of [HLQ24, Proposition 5.1] need not be T-invariant. In [HLQ24, Proof of Proposition 5.1], which follows [Zhu24a, Proof of Theorem 4.8], the enlarged boundary is klt by Bertini's theorem for a general member of a free linear system, while the T-invariant members come from T-semi-invariant sections and are not general once two weights occur. A component s_α of s is T-semi-invariant, but $w(B^+ - B) \leq w(G)$ may fail for valuations $w \neq \mathrm{wt}_\xi$. For s_α of minimal ξ-weight, the bound at wt_ξ is kept, and a bound at a minimizer of the normalized volume implies log canonicity up to a factor c_0 depending only on d [Zhu24a, Lemma 3.4], while no such factor works for all valuations (Remark 4.2). Here the K-semistability of $(X \ni x, B; \xi)$ is used for the first time.

(b) Arbitrarily large multiples. Lemma 4.1 only implies that $(X \ni x, B + c_0\lambda(B^+ - B))$ is lc, so Proposition 4.3 needs $c_0\lambda > 1$. Moreover, $c_0 \leq \frac{1}{d}$ (Remark 4.2). In [Zhu24a, Proof of Theorem 4.8] and [HLQ24, Proof of Proposition 5.1], the pair stays klt after the added divisor is multiplied by some $\mu \leq \frac{3}{2}$ and some $\mu < 2$, respectively, which does not suffice when $d \geq 2$, since $c_0\lambda > 1$ and $c_0 \leq \frac{1}{d}$ imply $\lambda > d \geq 2$. In Proposition 3.1, $\bar{B} = \frac{1}{q}\lfloor q(1+\tau)B \rfloor$ with τ proportional to $\frac{1}{\lambda}$. Since $\bar{B} \leq (1+\tau)B$ and $\lambda\tau$ depends only on d and ϵ, [Zhu24a, Theorem 1.3] implies that $(Y, \bar{B}_Y + \lambda(\bar{B}_Y - B_Y))$ is klt, where Y is a small $\mathbb{Q}$-factorial modification of a neighborhood of x [BCHM10, Theorem 1.2(2) and Corollary 1.4.3] and $B_Y, \bar{B}_Y$ are the strict transforms of $B, \bar{B}$. Then [HLQ24, Lemma 2.6] and [Fuj09, Theorem 2.2.4] make $|-N(K_Y + \bar{B}_Y)|$ base point free, and a general member is added with coefficient $\frac{\lambda}{N} < 1$.

(c) Volume ratios. Enlarging the boundary can move the minimizer of the normalized volume: $(X \ni x, B^+; \xi')$ need not be K-semistable for any Reeb vector ξ' of T (Remark 5.2). Since NB^+ is a Weil divisor and (X, B^+) is klt, $B^+ \in \{\frac{j}{N} \mid 0 \leq j \leq N-1\}$, and instead of [XZ24, Theorem 1.1], Proposition 5.1 uses [XZ24, Theorem 1.4], which replaces K-semistability by a lower bound of the volume ratio $\Theta(X, B^+; \xi)$ (Definition 2.5). By Lemma 2.3 and the upper bound d^d for the local volume of a d-dimensional klt singularity [LX19, Lemma A.1] (see [Zhu24b, Lemma 2.6] for $\mathbb{R}$-divisors),

$$\widehat{\mathrm{vol}}_{X,B^+}(\mathrm{wt}_\xi) \leq \widehat{\mathrm{vol}}_{X,B}(\mathrm{wt}_\xi) = \widehat{\mathrm{vol}}(X \ni x, B) \leq d^d,$$

so $\Theta(X, B^+; \xi) \geq 2^{-d}\epsilon/d^d$. The equality uses the K-semistability of $(X \ni x, B; \xi)$ for the second and last time. In [XZ24, p. 2], volume ratios allow perturbations of the polarization, and here they allow enlarging the boundary. By [XZ24, Theorem 1.4], the triples $(X, \operatorname{Supp} B^+; x)$ are log bounded, and Lemma 5.3 removes the irreducible components of $\operatorname{Supp} B^+$ not contained in $\operatorname{Supp} B$.

Finally, the lower bound η is used only to choose q in $\bar{B}$, so that the rounding error, less than $\frac{1}{q}$, is small compared with the coefficients of B. No step requires ξ or B to be rational: a component of minimal ξ-weight exists for any Reeb vector, real coefficients are handled by Lemma 4.1 and [HLQ24, Theorem 2.5 and Lemma 2.4], and B^+ is a $\mathbb{Q}$-divisor.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce log Fano cone singularities, their volume ratios, and log bounded sets of triples with marked points.

In Section 3, for klt singularities whose local volumes and boundary coefficients are bounded away from zero, we enlarge the boundary to a boundary with bounded denominators given by a regular function (Proposition 3.1).

In Section 4, for K-semistable log Fano cone singularities, we replace such a boundary with a torus-invariant one (Lemma 4.1 and Proposition 4.3).

In Section 5 we prove Lemma 5.3 and Proposition 5.1, a boundedness criterion in terms of such torus-invariant boundaries.

In Section 6 we prove Theorem 1.2 and Corollary 1.3.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and local volumes.

Definition 2.1. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X. We say that a pair $(X \ni x, B)$ is *lc* (resp. *klt*) if $(U, B|_U)$ is lc (resp. klt) for some open neighborhood U of x.

Definition-Lemma 2.2 (cf. [XZ24, Definition 2.1])**.** Let $(X \ni x, B)$ be a pair of dimension d. We denote by $\operatorname{Val}_{X,x}$ the set of real valuations of the function field of X that are centered at x. For any $v \in \operatorname{Val}_{X,x}$, we denote by $a(v, X, B) \in \mathbb{R} \cup \{+\infty\}$ the log discrepancy of v with respect to (X, B) ([JM12], [Blu18, Section 7.1], and [XZ24, Section 2.2]). For any $v \in \operatorname{Val}_{X,x}$ and any

$\mathbb{R}$-Cartier $\mathbb{R}$-divisor F on X, we write $F = \sum_j r_j \operatorname{div}(h_j)$ near x, where r_j are real numbers and h_j are rational functions on X, and let $v(F) := \sum_j r_j v(h_j)$. The number $v(F)$ does not depend on the choice of the r_j and h_j. The *volume* of $v \in \operatorname{Val}_{X,x}$ is

$$\operatorname{vol}(v) := \limsup_{m\to+\infty} \frac{\ell(\mathcal{O}_{X,x}/\mathfrak{a}_m(v))}{m^d/d!},$$

where $\mathfrak{a}_m(v) := \{f \in \mathcal{O}_{X,x} \mid v(f) \geq m\}$. Assume that $(X \ni x, B)$ is klt. The *normalized volume* of $v \in \operatorname{Val}_{X,x}$ is $\widehat{\operatorname{vol}}_{X,B}(v) := a(v, X, B)^d \operatorname{vol}(v)$ if $a(v, X, B) < +\infty$, and $\widehat{\operatorname{vol}}_{X,B}(v) := +\infty$ otherwise. The *local volume* of $(X \ni x, B)$ is

$$\widehat{\operatorname{vol}}(X \ni x, B) := \inf_{v\in\operatorname{Val}_{X,x}} \widehat{\operatorname{vol}}_{X,B}(v).$$

Proof. Suppose that

$$F = \sum_j r_j \operatorname{div}(h_j) = \sum_j r'_j \operatorname{div}(h'_j)$$

on an open neighborhood U of x. Let $g_1, \ldots, g_n$ be the distinct rational functions among the h_j and the h'_j. Then there exists $c = (c_1, \ldots, c_n) \in \mathbb{R}^n$ such that

$$\sum_j r_j v(h_j) - \sum_j r'_j v(h'_j) = \sum_{i=1}^n c_i v(g_i) \quad \text{and} \quad \sum_{i=1}^n c_i \operatorname{div}(g_i)|_U = 0.$$

Let $V \subset \mathbb{R}^n$ be the set of real solutions c' of the equation $\sum_i c'_i \operatorname{div}(g_i)|_U = 0$. Comparing the coefficients of the prime divisors on U, we see that V is the solution space of a homogeneous system of linear equations with integer coefficients. Thus $c = \sum_{l=1}^k \lambda_l c^{(l)}$ for some $c^{(1)}, \ldots, c^{(k)} \in V \cap \mathbb{Z}^n$ and some real numbers λ_l. For any l, the rational function $u_l := \prod_i g_i^{c_i^{(l)}}$ satisfies $\operatorname{div}(u_l)|_U = 0$. Since U is normal, u_l is a unit of $\mathcal{O}_{X,x}$, and $v(u_l) = 0$ because v is centered at x. Therefore

$$\sum_{i=1}^n c_i v(g_i) = \sum_{l=1}^k \lambda_l \sum_{i=1}^n c_i^{(l)} v(g_i) = \sum_{l=1}^k \lambda_l v(u_l) = 0.$$

□

Lemma 2.3. *Let $(X \ni x, B)$ be a pair and $v \in \operatorname{Val}_{X,x}$. Then:*

(1) For any $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $F \geq 0$ on X, we have $v(F) \geq 0$ and $a(v, X, B + F) = a(v, X, B) - v(F)$.

(2) For any pair $(X \ni x, B')$ and any real number $t \in [0, 1]$, we have

$$a(v, X, tB + (1 - t)B') = ta(v, X, B) + (1 - t)a(v, X, B').$$

(3) If $(X \ni x, B)$ is lc, then $a(v, X, B) \geq 0$.

In particular, if $F \geq 0$ is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X and $(X \ni x, B+F)$ is klt, then $(X \ni x, B)$ is klt and $\widehat{\operatorname{vol}}(X \ni x, B) \geq \widehat{\operatorname{vol}}(X \ni x, B + F)$.

Proof. Let F be an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X such that $B + F \geq 0$. Possibly shrinking X to an affine open neighborhood of x, we may assume that $K_X + B$ is $\mathbb{R}$-Cartier and $F = \sum_{j=1}^k r_j \operatorname{div}(f_j)$ for some real numbers r_j and rational functions f_j on X. Let $h : Y \to X$ be a log resolution of $(X, \operatorname{Supp} B \cup \operatorname{Supp} F)$, write $K_Y + B_Y := h^*(K_X + B)$, and let A_Y be the log discrepancy function of the smooth variety Y on real valuations [JM12, Section 5.2]. Since h is proper, v has a center on Y, and $a(v, X, B) = A_Y(v) - v(B_Y)$ (cf. [Blu18, Section 7.1] and [XZ24, Section 2.2]). Since $K_Y + B_Y + h^*F = h^*(K_X + B + F)$ and $v(h^*F) = \sum_j r_j v(f_j) = v(F)$, we obtain

$$a(v, X, B + F) = A_Y(v) - v(B_Y) - v(F) = a(v, X, B) - v(F), \tag{2.1}$$

where both sides are $+\infty$ if $A_Y(v) = +\infty$, because $v(B_Y)$ and $v(F)$ are real numbers.

(1) Assume that $F \geq 0$. Let $\sigma \subset \mathbb{R}^k$ be the set of vectors (t_j) such that $\sum_j t_j \operatorname{div}(f_j) \geq 0$. Then σ is a rational polyhedral cone, defined by one linear inequality with integer coefficients for

each prime divisor in $\bigcup_j \operatorname{Supp}\operatorname{div}(f_j)$, and $(r_j) \in \sigma$. Since a rational polyhedral cone is generated by finitely many integral vectors, we have $(r_j) = \sum_l \mu_l t^{(l)}$ for some real numbers $\mu_l \geq 0$ and some $t^{(l)} \in \sigma \cap \mathbb{Z}^k$. For any l, the rational function $u_l := \prod_j f_j^{t_j^{(l)}}$ satisfies $\operatorname{div}(u_l) \geq 0$, so u_l is a regular function on the normal affine variety X, and

$$F = \sum_l \mu_l \operatorname{div}(u_l). \tag{2.2}$$

Since v is centered at x, $v(u_l) \geq 0$ for any l. Thus $v(F) = \sum_l \mu_l v(u_l) \geq 0$. The equality in (1) is (2.1).

(2) Since $K_X + B$ and $K_X + B'$ are $\mathbb{R}$-Cartier near x, $B' - B$ is $\mathbb{R}$-Cartier near x. Applying (2.1) to $F = B' - B$ and to $F = (1-t)(B' - B)$, we have

$$\begin{aligned} a(v, X, B') &= a(v, X, B) - v(B' - B), \\ a(v, X, tB + (1-t)B') &= a(v, X, B) - (1-t)v(B' - B). \end{aligned} \tag{2.3}$$

Since $v(B' - B)$ is a real number, $a(v, X, B)$ and $a(v, X, B')$ are both finite or both $+\infty$. In the first case, (2) follows from (2.3). In the second case, $a(v, X, tB + (1-t)B') = +\infty$ and both sides of (2) are $+\infty$, with the convention $0 \cdot (+\infty) = 0$.

(3) Possibly shrinking X to an open neighborhood of x, we may assume that (X, B) is lc. Let $h : Y \to X$ be a log resolution of (X, B), write $K_Y + B_Y := h^*(K_X + B)$, and let $D = \sum_{i=1}^m D_i$ be the reduced divisor with $\operatorname{Supp} D = \operatorname{Supp} B_Y \cup \operatorname{Exc}(h)$, which is a simple normal crossing divisor. Then

$$B_Y = \sum_{i=1}^m (1 - a(D_i, X, B))D_i.$$

Let $\operatorname{QM}(Y, D)$ be the set of quasi-monomial valuations on Y with respect to (Y, D), and $r_{Y,D}$ the retraction onto $\operatorname{QM}(Y, D)$ [JM12, Section 4.3]. Then $r_{Y,D}(v)(D_i) = v(D_i)$ for any i. If $A_Y(v) = +\infty$, then $a(v, X, B) = +\infty$. Otherwise, by [JM12, Corollary 5.4], $A_Y(v) \geq A_Y(r_{Y,D}(v))$, and by [JM12, Proposition 5.1], $A_Y(r_{Y,D}(v)) = \sum_i r_{Y,D}(v)(D_i) = \sum_i v(D_i)$. Since v has a center on Y, $v(D_i) \geq 0$ for any i. Thus

$$\begin{aligned} a(v, X, B) = A_Y(v) - v(B_Y) &\geq \sum_{i=1}^m v(D_i) - \sum_{i=1}^m (1 - a(D_i, X, B))v(D_i) \\ &= \sum_{i=1}^m v(D_i)a(D_i, X, B) \geq 0. \end{aligned}$$

Finally, let $F \geq 0$ be an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X such that $(X \ni x, B + F)$ is klt. Possibly shrinking X to an affine open neighborhood of x, we may assume that $(X, B + F)$ is klt and (2.2) holds with regular functions u_l on X and real numbers $\mu_l \geq 0$. Let E be a prime divisor over X, and $h : Y \to X$ a log resolution of $(X, \operatorname{Supp} B \cup \operatorname{Supp} F)$ such that E is a divisor on Y. Since $K_Y + B_Y + h^*F = h^*(K_X + B + F)$, we have

$$\begin{aligned} a(E, X, B) &= a(E, X, B + F) + \operatorname{mult}_E h^*F \\ &= a(E, X, B + F) + \sum_l \mu_l \operatorname{mult}_E h^* \operatorname{div}(u_l) \geq a(E, X, B + F) > 0 \end{aligned}$$

(cf. [KM98, Lemma 2.27]). Thus (X, B) is klt. For any $v \in \operatorname{Val}_{X,x}$, (1) and (3) imply that $a(v, X, B) \geq a(v, X, B + F) \geq 0$. Hence $\widehat{\operatorname{vol}}_{X,B}(v) \geq \widehat{\operatorname{vol}}_{X,B+F}(v)$, also when $a(v, X, B) = +\infty$. Taking the infimum over $v \in \operatorname{Val}_{X,x}$, we obtain $\widehat{\operatorname{vol}}(X \ni x, B) \geq \widehat{\operatorname{vol}}(X \ni x, B + F)$. □

2.2. **Log Fano cone singularities.**

Definition 2.4 ([XZ24, Definitions 2.2 and 2.3])**.** Let $X = \operatorname{Spec} R$ be a normal affine variety and T an algebraic torus acting on X. We say that the T-action on X is *good* if it is faithful and there exists a unique closed point $x \in X$ that is contained in the closure of any T-orbit. In this case, we call x the *vertex* of X. Let $M := \operatorname{Hom}(T, \mathbb{G}_m)$ be the weight lattice of T, and let $R = \bigoplus_{\alpha \in M} R_\alpha$ be the weight decomposition of R. For any $s \in R$ and any $\alpha \in M$, we denote by $s_\alpha \in R_\alpha$ the component of s of weight α. A *Reeb vector* on X is an element $\xi \in \operatorname{Hom}(M, \mathbb{R})$ such that $\langle \xi, \alpha \rangle > 0$ for any $0 \neq \alpha \in M$ with $R_\alpha \neq 0$. For any Reeb vector ξ on X, we denote by $\mathrm{wt}_\xi \in \mathrm{Val}_{X,x}$ the valuation defined by

$$\mathrm{wt}_\xi(s) := \min\{\langle \xi, \alpha \rangle \mid \alpha \in M,\ s_\alpha \neq 0\}$$

for any $0 \neq s \in R$ (cf. [XZ24, Section 2.3]).

Definition 2.5 ([XZ24, Definitions 2.4, 2.5, and 2.7])**.** Let $(X \ni x, B)$ be a pair such that $X = \operatorname{Spec} R$ is affine. We say that $(X \ni x, B)$ is a *log Fano cone singularity* if $(X \ni x, B)$ is klt and there exist a non-trivial algebraic torus T and a good T-action on X such that x is the vertex of X and B is T-invariant. A *polarized log Fano cone singularity* $(X \ni x, B; \xi)$ consists of a log Fano cone singularity $(X \ni x, B)$ together with such a T-action and a Reeb vector ξ on X, and we call T the *acting torus* of $(X \ni x, B; \xi)$. We say that $(X \ni x, B; \xi)$ is *K-semistable* if $\widehat{\mathrm{vol}}(X \ni x, B) = \widehat{\mathrm{vol}}_{X,B}(\mathrm{wt}_\xi)$. We say that $(X \ni x, B)$ is a *K-semistable log Fano cone singularity* if there exist a good T-action on X and a Reeb vector ξ on X with respect to this T-action such that $(X \ni x, B; \xi)$ is a K-semistable polarized log Fano cone singularity. The *volume ratio* of $(X \ni x, B; \xi)$ is

$$\Theta(X, B; \xi) := \frac{\widehat{\mathrm{vol}}(X \ni x, B)}{\widehat{\mathrm{vol}}_{X,B}(\mathrm{wt}_\xi)}.$$

We follow [XZ24, Definition 2.5], where B is an $\mathbb{R}$-divisor (see [XZ24, Section 2.1]), and [XZ24, Theorem 2.6] recalled in the introduction holds for $\mathbb{R}$-divisors B with K-semistability as in Definition 2.5 (see [Zhu24b, Theorem 2.19]). K-semistability of polarized Fano cone singularities was introduced in [CS18, CS19]. When B is a $\mathbb{Q}$-divisor, the K-semistability in Definition 2.5 coincides with the K-semistability defined through special test configurations by [LX18, Theorem 2.34]. Section 5 uses the volume ratio to handle polarized log Fano cone singularities obtained by enlarging the boundary, which need not be K-semistable.

Lemma 2.6. *Let $X = \operatorname{Spec} R$ be a normal affine variety with a good action of an algebraic torus T, and x the vertex of X. Then:*

1. *Any non-empty closed T-invariant subset of X contains x.*
2. *For any T-invariant $\mathbb{R}$-divisor $B \geq 0$ on X such that $(X \ni x, B)$ is a klt pair, (X, B) is a klt pair.*
3. *For any T-invariant Weil divisor $F \geq 0$ on X, the ideal $H^0(X, \mathcal{O}_X(-F)) \subset R$ is homogeneous, that is, $s_\alpha \in H^0(X, \mathcal{O}_X(-F))$ for any $s \in H^0(X, \mathcal{O}_X(-F))$ and any $\alpha \in M$.*
4. *There exists a T-invariant canonical divisor K_X of X such that $K_X \geq 0$.*

Proof. (1) Let V be a non-empty closed T-invariant subset of X and $y \in V$. Then V contains the closure of $T \cdot y$, which contains x.

(2) Let U be an open neighborhood of x such that $K_U + B|_U$ is $\mathbb{R}$-Cartier and $(U, B|_U)$ is klt. For any $t \in T$, t is an automorphism of X such that $t_* B = B$ and $t_* K_X \sim K_X$. Thus $K_X + B$ is $\mathbb{R}$-Cartier on tU, and $(tU, B|_{tU})$ is klt. Hence $K_X + B$ is $\mathbb{R}$-Cartier on $U' := \bigcup_{t \in T} tU$, and $(U', B|_{U'})$ is klt. Since U' is a T-invariant open subset of X containing x, $X \backslash U'$ is a closed T-invariant subset of X not containing x, so $X \backslash U' = \emptyset$ by (1).

(3) Let $\mathfrak{a} := H^0(X, \mathcal{O}_X(-F))$. For any $t \in T$ and any $f \in \mathfrak{a}$, we have

$$\operatorname{div}(t^* f) = t^* \operatorname{div}(f) \geq t^* F = F,$$

so $t^*f \in \mathfrak{a}$. Thus $\mathfrak{a}$ is a T-stable subspace of R. The T-action on R induces the weight decomposition $R = \bigoplus_{\alpha\in M} R_\alpha$, and

$$\mathfrak{a} = \bigoplus_{\alpha\in M} (\mathfrak{a} \cap R_\alpha)$$

because distinct characters of T are linearly independent as functions on T.

(4) Let $j : X_{\mathrm{reg}} \to X$ be the inclusion and $\omega_X := j_*\Omega^{\dim X}_{X_{\mathrm{reg}}}$. The group T acts on $\Omega^{\dim X}_{X_{\mathrm{reg}}}$ by pullback, which induces a T-linearization of ω_X. Thus $H^0(X, \omega_X)$ has a coaction

$$H^0(X, \omega_X) \to H^0(X, \omega_X) \otimes \mathbb{C}[M]$$

and a weight decomposition. Since ω_X is a torsion-free sheaf of rank one on the affine variety X, $H^0(X, \omega_X) \neq 0$. Let $\omega \in H^0(X, \omega_X)$ be a non-zero homogeneous element. Then $K_X := \mathrm{div}(\omega) \geq 0$ is a T-invariant canonical divisor of X. □

2.3. **Log bounded families.**

Definition 2.7. Let $\mathcal{C}$ be a set of triples $(X, D; x)$, where X is a variety, $D \subset X$ is a closed subset, and $x \in X$ is a closed point. We say that $\mathcal{C}$ is *log bounded* if there exist a morphism $f : \mathcal{X} \to S$ between schemes of finite type over $\mathbb{C}$, a closed subset $\mathcal{D} \subset \mathcal{X}$, and a section $\sigma : S \to \mathcal{X}$ of f, such that for any $(X, D; x) \in \mathcal{C}$, there exist a closed point $t \in S$ and an isomorphism $g : X \to \mathcal{X}_t := f^{-1}(t)$ such that $g(D) = \mathcal{D} \cap \mathcal{X}_t$ and $g(x) = \sigma(t)$.

Definition 2.7 only keeps track of the closed subset D and the marked point x, not of coefficients: in Theorem 1.2 we take $D = \operatorname{Supp} B$, and the coefficients of B may take infinitely many real values.

3. Boundaries with bounded denominators

In this section, we prove Proposition 3.1, which is step (1) of the sketch of the proof in Section 1.

Proposition 3.1 (cf. [HLQ24, Proposition 5.1])**.** *Let d be a positive integer and ϵ, η, λ three positive real numbers such that $\lambda > 1$. Then there exist two positive real numbers τ, δ and two positive integers q, N depending only on $d, \epsilon, \eta, \lambda$ such that $q \mid N$, $N > \lambda$, and the following holds.*

Let (X, B) be a klt pair of dimension d, $x \in X$ a closed point, $K_X \geq 0$ a canonical divisor of X, and $\bar{B} := \frac{1}{q}\lfloor q(1+\tau)B \rfloor$. Assume that X is affine, $\widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon$, and $\mathrm{mult}_P B \geq \eta$ for any prime divisor $P \subset \operatorname{Supp} B$. Then:

(1) $\operatorname{Supp} \bar{B} = \operatorname{Supp} B$ and $(1+\delta)B \leq \bar{B} \leq (1+\tau)B$.

(2) There exists a non-zero element

$$s \in H^0(X, \mathcal{O}_X(-N(K_X + \bar{B}))) \subset H^0(X, \mathcal{O}_X)$$

such that the $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $G := \frac{1}{N}\mathrm{div}(s) - (K_X + B)$ satisfies $G \geq \delta B$ and $(X, B + \lambda G)$ is klt near x.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we choose τ, δ, q, and prove (1).

Let $c(d) > 0$ be the constant of [Zhu24a, Theorem 1.3]: for any d-dimensional klt pair $(Y \ni y, \Delta)$ such that K_Y is $\mathbb{Q}$-Cartier, the pair $(Y \ni y, (1+t)\Delta)$ is lc for any real number t such that $0 \leq t \leq c(d)\widehat{\mathrm{vol}}(Y \ni y, \Delta)$. The theorem holds for $\mathbb{R}$-divisors by [HLQ24, Theorem 2.5] and the paragraph before [HLQ24, Lemma 2.3]. Let

$$\gamma := \frac{1}{2}\min\{1, c(d)\epsilon\}, \quad \tau := \frac{\gamma}{2\lambda}, \quad \delta := \frac{\tau}{2},$$

and let q be a positive integer such that $q \geq \frac{2}{\tau\eta}$. Then τ, δ, q depend only on $d, \epsilon, \eta, \lambda$.

Let P be a prime divisor on X, $b := \mathrm{mult}_P B$, and $\bar{b} := \mathrm{mult}_P \bar{B} = \frac{1}{q}\lfloor q(1+\tau)b \rfloor$. If $b = 0$, then $\bar{b} = 0$. If $b > 0$, then $b \geq \eta$, so $\frac{1}{q} \leq \frac{\tau\eta}{2} \leq \frac{\tau b}{2}$, and

$$(1+\delta)b = (1+\tau)b - \frac{\tau b}{2} \leq (1+\tau)b - \frac{1}{q} < \bar{b} \leq (1+\tau)b.$$

In particular, $\bar{b} > 0$. Thus (1) holds.

Step 2. In this step, we construct an affine open neighborhood U of x and a small birational morphism $f : Y \to U$ such that Y is $\mathbb{Q}$-factorial and $-(K_Y + \bar{B}_Y)$ is nef over U. Since Y is $\mathbb{Q}$-factorial, the divisors on Y that we use are $\mathbb{R}$-Cartier, and $\mathbb{Q}$-Cartier when their coefficients are rational. In particular, Y is $\mathbb{Q}$-Gorenstein and B_Y is $\mathbb{R}$-Cartier, which are the hypotheses of [Zhu24a, Theorem 1.3] on Y, while B itself need not be $\mathbb{R}$-Cartier. The nefness of $-(K_Y + \bar{B}_Y)$ over U is the condition of [Fuj09, Theorem 2.2.4] in **Step 3**.

Let $U \subset X$ be an affine open neighborhood of x, $K_U := K_X|_U$, $B_U := B|_U$, and $\bar{B}_U := \bar{B}|_U$. For any normal variety Y_1 with a small birational morphism $f_1 : Y_1 \to U$, let $B_{Y_1} := f_{1*}^{-1} B_U$, and similarly write the strict transform f_{1*}^{-1} on Y_1 of any other $\mathbb{R}$-divisor on U with the subscript Y_1. Then K_{Y_1} is a canonical divisor of Y_1, and

$$K_{Y_1} + B_{Y_1} = f_1^*(K_U + B_U),$$

since both sides coincide outside the exceptional locus of f_1, which has codimension at least 2. Hence $a(E, Y_1, B_{Y_1}) = a(E, U, B_U)$ for any prime divisor E over U, so (Y_1, B_{Y_1}) is klt [KM98, Lemma 2.30] ([KM98, Lemma 2.30] is formulated for $\mathbb{Q}$-divisors, and its proof works for $\mathbb{R}$-divisors without change).

Since U is affine, there exists a non-zero regular function u on U such that

$$D' := \frac{1}{q}\,\mathrm{div}_U(u) - (K_U + \bar{B}_U) \geq 0$$

(take u vanishing to a large order along the prime divisors in $\mathrm{Supp}(K_U + \bar{B}_U)$). By [BCHM10, Corollary 1.4.3] and the paragraph following it, there exists a small $\mathbb{Q}$-factorialization $f_0 : Y_0 \to U$ of (U, B_U) (f_0 is projective, being a log terminal model over U [BCHM10, Definition 3.6.7]). Then (Y_0, B_{Y_0}) is klt and $D'_{Y_0} \geq 0$ is $\mathbb{Q}$-Cartier, so $(Y_0, B_{Y_0} + tD'_{Y_0})$ is klt for some rational number $t > 0$, as one checks on a log resolution of $(Y_0, B_{Y_0} + D'_{Y_0})$ (cf. [KM98, Corollary 2.32]). Since f_0 is birational, $K_{Y_0} + B_{Y_0} + tD'_{Y_0} = f_0^*(K_U + B_U) + tD'_{Y_0}$ is big over U. By [BCHM10, Theorem 1.2(2)], $(Y_0, B_{Y_0} + tD'_{Y_0})$ has a log canonical model $g : Y' \to U$ over U. The birational map $Y_0 \dashrightarrow Y'$ does not extract any divisor and f_0 is small, so g is small. Thus $K_{Y'} + B_{Y'} = g^*(K_U + B_U)$, $(Y', B_{Y'})$ is klt, and

$$tD'_{Y'} = (K_{Y'} + B_{Y'} + tD'_{Y'}) - g^*(K_U + B_U)$$

is $\mathbb{R}$-Cartier and ample over U. Since $D'_{Y'}$ has rational coefficients, $D'_{Y'}$ is $\mathbb{Q}$-Cartier.

By [BCHM10, Corollary 1.4.3] and the paragraph following it again, there exists a small $\mathbb{Q}$-factorialization $p : Y \to Y'$ of $(Y', B_{Y'})$ (p is projective, being a log terminal model over Y' [BCHM10, Definition 3.6.7]). Let $f := g \circ p$. Then f is small, Y is $\mathbb{Q}$-factorial, $K_Y = f_*^{-1} K_U$, $B_Y = f_*^{-1} B$, and $\bar{B}_Y = f_*^{-1} \bar{B}$. Since p is small and $D'_{Y'}$ is $\mathbb{Q}$-Cartier, $D'_Y = p^* D'_{Y'}$. Since f is small, the strict transform of $\mathrm{div}_U(u)$ on Y is $\mathrm{div}_Y(u)$. Thus

$$-(K_Y + \bar{B}_Y) = p^* D'_{Y'} - \frac{1}{q}\,\mathrm{div}_Y(u)$$

is the sum of the pullback of a g-ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor and $-\frac{1}{q}$ times a principal divisor. Hence $-(K_Y + \bar{B}_Y)$ is nef over U.

Step 3. In this step, possibly shrinking U, we bound the singularities of $(Y, \bar{B}_Y)$ and the Cartier index of $q(K_Y + \bar{B}_Y)$ uniformly, and define N.

Let $y \in f^{-1}(x)$ be a closed point. Since $K_Y + B_Y = f^*(K_U + B_U)$ and (Y, B_Y) is klt (**Step 2**), by [Zhu24a, Lemma 2.10] (see [HLQ24, Lemma 2.4] for $\mathbb{R}$-divisors),

$$\widehat{\mathrm{vol}}(Y \ni y, B_Y) \geq \widehat{\mathrm{vol}}(X \ni x, B) \geq \epsilon.$$

Since Y is $\mathbb{Q}$-factorial, K_Y is $\mathbb{Q}$-Cartier. Since $2\gamma \leq c(d)\epsilon$, by the choice of $c(d)$ in **Step 1**, $(Y \ni y, (1+2\gamma)B_Y)$ is lc. For any prime divisor E over a neighborhood of y on which (Y, B_Y) is klt and $(Y, (1+2\gamma)B_Y)$ is lc,

$$a(E, Y, (1+\gamma)B_Y) = \frac{1}{2}a(E, Y, B_Y) + \frac{1}{2}a(E, Y, (1+2\gamma)B_Y) > 0$$

(cf. Lemma 2.3(2)), so $(Y \ni y, (1+\gamma)B_Y)$ is klt. For any $v \in \mathrm{Val}_{Y,y}$, Lemma 2.3(2)(3) implies $a(v, Y, (1+\gamma)B_Y) \geq \frac{1}{2}a(v, Y, B_Y) \geq 0$, so $\widehat{\mathrm{vol}}_{Y,(1+\gamma)B_Y}(v) \geq 2^{-d}\widehat{\mathrm{vol}}_{Y,B_Y}(v)$. Thus

$$\widehat{\mathrm{vol}}(Y \ni y, (1+\gamma)B_Y) \geq 2^{-d}\widehat{\mathrm{vol}}(Y \ni y, B_Y) \geq 2^{-d}\epsilon$$

(cf. [HLQ23, Lemma 2.16] and [Zhu24a, Lemma 2.11]).

Let $\Delta_Y := B_Y + \lambda(\bar{B}_Y - B_Y)$. By (1), $B_Y \leq \bar{B}_Y \leq (1+\tau)B_Y$. Since $\lambda\tau = \frac{\gamma}{2}$ and $\tau < \gamma$, we have

$$B_Y \leq \Delta_Y \leq \left(1 + \frac{\gamma}{2}\right)B_Y \leq (1+\gamma)B_Y \quad \text{and} \quad B_Y \leq \bar{B}_Y \leq (1+\gamma)B_Y.$$

Since Y is $\mathbb{Q}$-factorial, $(1+\gamma)B_Y - \Delta_Y \geq 0$ and $(1+\gamma)B_Y - \bar{B}_Y \geq 0$ are $\mathbb{R}$-Cartier. By Lemma 2.3, $(Y \ni y, \Delta_Y)$ and $(Y \ni y, \bar{B}_Y)$ are klt, and

$$\widehat{\mathrm{vol}}(Y \ni y, \bar{B}_Y) \geq \widehat{\mathrm{vol}}(Y \ni y, (1+\gamma)B_Y) \geq 2^{-d}\epsilon.$$

Possibly shrinking U to an affine neighborhood of x, we may assume that (Y, Δ_Y) and $(Y, \bar{B}_Y)$ are klt (cf. [KM98, Corollary 2.32]).

Since $q\bar{B} = \lfloor q(1+\tau)B \rfloor$, $q(K_Y + \bar{B}_Y)$ is a Weil divisor, and it is $\mathbb{Q}$-Cartier since Y is $\mathbb{Q}$-factorial. Let V be an irreducible component of $f^{-1}(x)$. Then V is a proper closed subvariety of Y, and $\widehat{\mathrm{vol}}(Y \ni y, \bar{B}_Y) \geq 2^{-d}\epsilon$ for any closed point $y \in V$. By [HLQ24, Lemma 2.6] applied to the pair $(Y, \bar{B}_Y)$, the subvariety V, and the Weil divisor $q(K_Y + \bar{B}_Y)$, there exists a positive integer J depending only on d and $2^{-d}\epsilon$ such that $Jq(K_Y + \bar{B}_Y)$ is Cartier near any point of V. Since J does not depend on V and $f^{-1}(x)$ has finitely many irreducible components, $Jq(K_Y + \bar{B}_Y)$ is Cartier near $f^{-1}(x)$. Possibly shrinking U to an affine neighborhood of x again, we may assume that $Jq(K_Y + \bar{B}_Y)$ is Cartier.

Let $L := -Jq(K_Y + \bar{B}_Y)$. By **Step 2**, L is a Cartier divisor which is nef over U, and

$$L - (K_Y + \bar{B}_Y) = -(Jq+1)(K_Y + \bar{B}_Y)$$

is nef over U. Since f is birational, $L - (K_Y + \bar{B}_Y)$ is also big over U. Moreover, $(Y, \bar{B}_Y)$ is klt and $\bar{B}_Y$ has rational coefficients. By [Fuj09, Theorem 2.2.4], for an lc pair (Y, Δ) of dimension d with $\Delta \geq 0$ a $\mathbb{Q}$-divisor, a projective surjective morphism π from Y to a variety, and a π-nef Cartier divisor L such that $aL - (K_Y + \Delta)$ is π-nef and π-log big for some real number $a \geq 0$, the sheaf $\mathcal{O}_Y(mL)$ is π-generated for some positive integer m depending only on d and a. We apply it to $\Delta = \bar{B}_Y$, $\pi = f$ (note that $f = g \circ p$ is projective, since the log canonical model g and p are projective [BCHM10, Definition 3.6.7]), and $a = 1$. Since $(Y, \bar{B}_Y)$ is klt, log bigness over U is bigness over U. Thus there exists a positive integer m depending only on d such that $\mathcal{O}_Y(mL)$ is generated over U. Let

$$N := Jqm(\lceil \lambda \rceil + 1).$$

Then $q \mid N$, $N \geq \lceil \lambda \rceil + 1 > \lambda$, and N depends only on $d, \epsilon, \eta, \lambda$. Moreover,

$$\mathcal{O}_Y(-N(K_Y + \bar{B}_Y)) = \mathcal{O}_Y(mL)^{\otimes(\lceil \lambda \rceil + 1)}$$

is a line bundle generated over U.

Step 4. In this step, we take a general member D_Y of a free linear system contained in $|-N(K_Y + \bar{B}_Y)|$, show that $(Y, \Delta_Y + \frac{\lambda}{N}D_Y)$ is klt, and push it down to U.

Since U is affine and $\mathcal{O}_Y(-N(K_Y+\bar{B}_Y))$ is generated over U, it is generated by finitely many global sections $s_1,\dots,s_r$. Since Y is $\mathbb{Q}$-factorial and (Y,Δ_Y) is klt, there exists a $\mathbb{Q}$-divisor $\Delta'_Y \geq \Delta_Y$ such that $\operatorname{Supp}\Delta'_Y = \operatorname{Supp}\Delta_Y$ and (Y,Δ'_Y) is klt, by increasing the coefficients of Δ_Y slightly and checking on a log resolution of (Y,Δ_Y) (cf. [KM98, Corollary 2.32]). Let D_Y be the divisor of zeros of a general linear combination of $s_1,\dots,s_r$. Then D_Y is a general member of a free linear system. Thus D_Y is reduced by Bertini's theorem [Har77, III, Corollary 10.9 and Remark 10.9.2] applied on the smooth locus of Y, and, as the linear system is free, D_Y contains no component of Δ'_Y. Since (Y,Δ'_Y) is klt, [KM98, Lemma 5.17(2)] implies $a(E,Y,\Delta'_Y+D_Y)>0$ for any prime divisor E exceptional over Y. Since the coefficients of Δ'_Y are less than 1, (Y,Δ'_Y+D_Y) is lc. Since $\frac{\lambda}{N}<1$ and

$$\Delta'_Y+\frac{\lambda}{N}D_Y=\Big(1-\frac{\lambda}{N}\Big)\Delta'_Y+\frac{\lambda}{N}(\Delta'_Y+D_Y),$$

the pair $(Y,\Delta'_Y+\frac{\lambda}{N}D_Y)$ is klt (cf. Lemma 2.3(2)(3)). Since $\Delta'_Y-\Delta_Y\geq 0$ is $\mathbb{R}$-Cartier, $(Y,\Delta_Y+\frac{\lambda}{N}D_Y)$ is klt (cf. Lemma 2.3(1)).

We regard the general section as a rational function s_0 on Y, so that

$$D_Y=\operatorname{div}_Y(s_0)-N(K_Y+\bar{B}_Y).$$

Since f is small, $f_*D_Y=\operatorname{div}_U(s_0)-N(K_U+\bar{B}_U)\geq 0$. In particular, s_0 is a non-zero element of $H^0(U,\mathcal{O}_U(-N(K_U+\bar{B}_U)))$. Let

$$G_U:=\frac{1}{N}\operatorname{div}_U(s_0)-(K_U+B_U)=(\bar{B}_U-B_U)+\frac{1}{N}f_*D_Y.$$

Then $G_U\geq 0$ is $\mathbb{R}$-Cartier, since K_U+B_U is $\mathbb{R}$-Cartier. A direct calculation shows

$$\begin{aligned}K_Y+\Delta_Y+\frac{\lambda}{N}D_Y&=(1-\lambda)(K_Y+B_Y)+\frac{\lambda}{N}\operatorname{div}_Y(s_0)\\&=f^*\Big((1-\lambda)(K_U+B_U)+\frac{\lambda}{N}\operatorname{div}_U(s_0)\Big)=f^*(K_U+B_U+\lambda G_U),\end{aligned}$$

where the second equality follows from $K_Y+B_Y=f^*(K_U+B_U)$. Moreover,

$$f_*\Big(\Delta_Y+\frac{\lambda}{N}D_Y\Big)=B_U+\lambda G_U.$$

By [KM98, Lemma 2.30] (applied to $\mathbb{R}$-divisors as in **Step 2**), $(U,B_U+\lambda G_U)$ is klt, since f is small and $(Y,\Delta_Y+\frac{\lambda}{N}D_Y)$ is klt.

Step 5. In this step, we construct s. The section s_0 in **Step 4** is defined only on the neighborhood U of x. Multiplying it by a power of a regular function h, we obtain an element of the ideal $H^0(X,\mathcal{O}_X(-N(K_X+\bar{B})))$ of $H^0(X,\mathcal{O}_X)$, which is the form used in Section 4, while G coincides with G_U near x.

Since $K_X\geq 0$, $\bar{B}\geq 0$, and $q\mid N$, $N(K_X+\bar{B})\geq 0$ is a Weil divisor, so $\mathcal{O}_X(-N(K_X+\bar{B}))$ is a coherent ideal sheaf of $\mathcal{O}_X$. Since X is affine, there exists $h\in H^0(X,\mathcal{O}_X)$ such that $x\in D(h)\subset U$, where $D(h):=\{h\neq 0\}$. By [Har77, II, Proposition 5.1 and Corollary 5.5],

$$H^0(D(h),\mathcal{O}_X(-N(K_X+\bar{B})))=H^0(X,\mathcal{O}_X(-N(K_X+\bar{B})))_h.$$

Since $s_0|_{D(h)}$ belongs to the left-hand side, there exists a non-negative integer k such that

$$s:=h^k s_0\in H^0(X,\mathcal{O}_X(-N(K_X+\bar{B}))).$$

Then $s\neq 0$, and $G=\frac{1}{N}\operatorname{div}(s)-(K_X+B)$ is $\mathbb{R}$-Cartier. Since h is invertible on $D(h)$,

$$\operatorname{div}(s)|_{D(h)}=\operatorname{div}_U(s_0)|_{D(h)},$$

so $G|_{D(h)}=G_U|_{D(h)}$. By **Step 4**, $(X,B+\lambda G)$ is klt on $D(h)$, which is a neighborhood of x. Since $\operatorname{div}(s)-N(K_X+\bar{B})\geq 0$, by (1),

$$G=(\bar{B}-B)+\frac{1}{N}\big(\operatorname{div}(s)-N(K_X+\bar{B})\big)\geq\bar{B}-B\geq\delta B.$$

Thus (2) holds. □

Remark 3.2. Proposition 3.1 strengthens [HLQ24, Proposition 5.1] in two ways, both for Section 4. First, the multiple λ is arbitrary: after a homogeneous component of s is taken in Section 4, only the multiple $c_0\lambda$ is left, where c_0 is the constant in Lemma 4.1. Second, s is a regular function on X and the canonical divisor $K_X \geq 0$ is arbitrary: when a torus T acts on X, we take a T-invariant K_X, so the ideal $H^0(X, \mathcal{O}_X(-N(K_X + \bar{B})))$ is homogeneous and s decomposes into weight components. The boundary $\bar{B}$ alone does not suffice, since $K_X + \bar{B}$ need not be $\mathbb{R}$-Cartier (see the example $X = \{z_1z_2 = z_3z_4\}$ in Section 1). In the proof, λ enters through $\tau = \frac{\gamma}{2\lambda}$ in **Step 1** and through $N > \lambda$ in **Step 3**, so that the coefficient $\frac{\lambda}{N}$ of the general member in **Step 4** is less than 1. Otherwise, **Step 1**–**Step 4** follow [HLQ24, Proof of Proposition 5.1], and **Step 5** is new.

4. Torus-invariant boundaries with bounded denominators

In this section, we prove Proposition 4.3, which is step (2) of the sketch of the proof in Section 1. Replacing the section of Section 3 by a homogeneous component of minimal ξ-weight keeps the bound on the added divisor only at the valuation wt_ξ, and Lemma 4.1 turns a bound at a minimizer of the normalized volume into log canonicity.

Lemma 4.1 (cf. [Zhu24a, Lemma 3.4])**.** *Let d be a positive integer. Then there exists a real number $c_0 \in (0, 1]$ depending only on d satisfying the following. Let $(X \ni x, B)$ be a klt pair of dimension d and $D \geq 0$ an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X. Let $u \in \mathrm{Val}_{X,x}$ be a valuation such that $\widehat{\mathrm{vol}}_{X,B}(u) = \widehat{\mathrm{vol}}(X \ni x, B)$. Then $(X \ni x, B + tD)$ is lc for any real number $t \geq 0$ such that $t \cdot u(D) \leq c_0 \cdot a(u, X, B)$.*

Proof. Let $c_0(d) > 0$ be the real number given by [Zhu24a, Lemma 3.4] in dimension d, and let $c_0 := \min\{1, c_0(d)\}$. In our notation (the pair is lc at x for $0 \leq t \leq \mathrm{lct}(X \ni x, \Delta; D')$, which is $+\infty$ if $v_0(D') = 0$ because then $x \notin \operatorname{Supp} D'$, and the conventions of [Zhu24a, Section 2.1] that X is affine and $x \in \operatorname{Supp} \Delta$ whenever $\Delta \neq 0$ are achieved by shrinking X), [Zhu24a, Lemma 3.4] says that for any klt pair $(X \ni x, \Delta)$ of dimension d such that Δ is a $\mathbb{Q}$-divisor and $K_X + \Delta$ is $\mathbb{Q}$-Cartier, any $v_0 \in \mathrm{Val}_{X,x}$ with $a(v_0, X, \Delta) < +\infty$, and any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D' \geq 0$ on X, the pair $(X \ni x, \Delta + tD')$ is lc for any real number $t \geq 0$ such that

$$t \cdot v_0(D') \leq c_0(d) \frac{\widehat{\mathrm{vol}}(X \ni x, \Delta)}{\widehat{\mathrm{vol}}_{X,\Delta}(v_0)} a(v_0, X, \Delta).$$

Since [Zhu24a, Lemma 3.4] considers boundaries with rational coefficients and $\mathbb{Q}$-Cartier added divisors (see [Zhu24a, Section 2.1]), while B has real coefficients and D is only $\mathbb{R}$-Cartier, we approximate B and D by $\mathbb{Q}$-divisors. The proof is long, so we divide it into several steps.

Step 1. In this step we approximate B by $\mathbb{Q}$-divisors.

Since the hypotheses and the conclusion only depend on the germ of X at x and $K_X + B$ is $\mathbb{R}$-Cartier near x, possibly replacing X with an affine open neighborhood of x, we may assume that (X, B) is klt and

$$K_X + B = \sum_l r_l \operatorname{div}(\varphi_l)$$

for finitely many real numbers r_l and rational functions φ_l on X. Write $B = \sum_j b_j P_j$, where the P_j are distinct prime divisors and $b_j > 0$ for any j. Consider the equation

$$K_X + \sum_j y_j P_j = \sum_l z_l \operatorname{div}(\varphi_l)$$

in the unknowns (y_j, z_l). Comparing the coefficients of the finitely many prime divisors contained in the closed subset

$$\operatorname{Supp} K_X \cup \operatorname{Supp} B \cup \bigcup_l \operatorname{Supp} \operatorname{div}(\varphi_l),$$

we obtain a system of linear equations with integer coefficients and integer constant terms, and (b_j, r_l) is a real solution. Since the system has rational coefficients, Gaussian elimination over $\mathbb{Q}$ shows that the rational solutions are dense in the set of real solutions. Thus there exist rational solutions $(b_{k,j}, r_{k,l})$, $k = 1, 2, \ldots$, such that $b_{k,j} > 0$ for any k and j, and $(b_{k,j}, r_{k,l}) \to (b_j, r_l)$ as $k \to +\infty$. Let $B_k := \sum_j b_{k,j} P_j$. Then $B_k \geq 0$ is a $\mathbb{Q}$-divisor, $\operatorname{Supp} B_k = \operatorname{Supp} B$, and $K_X + B_k = \sum_l r_{k,l} \operatorname{div}(\varphi_l)$ is $\mathbb{Q}$-Cartier.

Step 2. In this step we compare the log discrepancies with respect to B and B_k, and the local volumes of $(X \ni x, B)$ and $(X \ni x, B_k)$. In general, u does not minimize $\widehat{\mathrm{vol}}_{X,B_k}$, so we use [Zhu24a, Lemma 3.4] in the form with the ratio $\widehat{\mathrm{vol}}(X \ni x, B_k)/\widehat{\mathrm{vol}}_{X,B_k}(u)$ rather than [Zhu24a, Corollary 3.5], and we bound $\widehat{\mathrm{vol}}(X \ni x, B_k)/\widehat{\mathrm{vol}}_{X,B_k}(u)$ from below by a quantity tending to 1.

Let $h : Y \to X$ be a log resolution of (X, B), let $E_1, \ldots, E_n$ be the prime divisors contained in $\operatorname{Exc}(h) \cup \operatorname{Supp} h_*^{-1} B$, and let $E := \sum_{i=1}^n E_i$. Write $K_Y + B_Y := h^*(K_X + B)$ and $K_Y + B_{k,Y} := h^*(K_X + B_k)$. Since $h_* B_Y = B$ and $h_* B_{k,Y} = B_k$, we have $\operatorname{Supp} B_Y \cup \operatorname{Supp} B_{k,Y} \subset \operatorname{Supp} E$. Since

$$B_{k,Y} - B_Y = \sum_l (r_{k,l} - r_l) h^* \operatorname{div}(\varphi_l),$$

we have

$$a(E_i, X, B_k) = 1 - \operatorname{mult}_{E_i} B_{k,Y} \to 1 - \operatorname{mult}_{E_i} B_Y = a(E_i, X, B) > 0$$

as $k \to +\infty$ for any i, where the last inequality holds because (X, B) is klt. Thus there exist real numbers $\delta_k \geq 0$ with $\delta_k \to 0$ such that

$$(1 - \delta_k) a(E_i, X, B) \leq a(E_i, X, B_k) \leq (1 + \delta_k) a(E_i, X, B)$$

for any i and k. Since $\delta_k \to 0$, possibly removing finitely many k, we may assume that $\delta_k < 1$ for any k.

Let w be a valuation of the function field of X with a center on X, and for any i, let $w(E_i)$ be the value of w on a local equation of E_i at the center of w on Y. Let A_Y be the log discrepancy function of the smooth variety Y on real valuations [JM12, Section 5.2]. By the definition of the log discrepancy of a valuation (cf. [Blu18, Section 7.1] and [XZ24, Section 2.2]), for $\Delta \in \{B, B_k\}$, we have

$$a(w, X, \Delta) = \Big(A_Y(w) - \sum_{i=1}^n w(E_i) \Big) + \sum_{i=1}^n a(E_i, X, \Delta) w(E_i).$$

Let $\mathrm{QM}(Y, E)$ be the set of quasi-monomial valuations on Y with respect to (Y, E), and $r_{Y,E}$ the retraction onto $\mathrm{QM}(Y, E)$ [JM12, Section 4.3]. Since Y is smooth, by [JM12, Proposition 5.1 and Corollary 5.4], $A_Y(w) \geq A_Y(r_{Y,E}(w)) = \sum_{i=1}^n w(E_i)$, that is, $A_Y(w) - \sum_{i=1}^n w(E_i) \geq 0$. Since $w(E_i) \geq 0$ for any i, we deduce that

$$(1 - \delta_k) a(w, X, B) \leq a(w, X, B_k) \leq (1 + \delta_k) a(w, X, B) \tag{4.1}$$

for any k and any valuation w of the function field of X with a center on X. In particular, $a(F, X, B_k) > 0$ for any prime divisor F over X, hence (X, B_k) is klt for any k.

We show that $0 < a(u, X, B) < +\infty$. Let $c := \min\{1, a(E_1, X, B), \ldots, a(E_n, X, B)\} > 0$. Since $A_Y(u) - \sum_{i=1}^n u(E_i) \geq 0$, we have

$$a(u, X, B) \geq c \Big(A_Y(u) - \sum_{i=1}^n u(E_i) + \sum_{i=1}^n u(E_i) \Big) = c \cdot A_Y(u).$$

Let y be the center of u on Y and z an element of a regular system of parameters of $\mathcal{O}_{Y,y}$. By [JM12, Proposition 5.1 and Corollary 5.4] for the smooth prime divisor $\{z = 0\}$ on a neighborhood of y (cf. [JM12, Remark 5.5]), we have $A_Y(u) \geq u(z) > 0$. Thus $a(u, X, B) > 0$. Moreover, $a(u, X, B) < +\infty$, since $\widehat{\mathrm{vol}}_{X,B}(u) = \widehat{\mathrm{vol}}(X \ni x, B) \leq \widehat{\mathrm{vol}}_{X,B}(v) < +\infty$ for any divisorial valuation $v \in \mathrm{Val}_{X,x}$ (such v exist, for instance the valuation defined by an irreducible component

of the exceptional divisor of the normalized blow-up of X at x). Here $\mathrm{vol}(v) \leq e(\mathfrak{m}_x)/v(\mathfrak{m}_x)^d < +\infty$, since for any positive integer m,

$$\mathfrak{a}_m(v) \supset \mathfrak{m}_x^{\lceil m/v(\mathfrak{m}_x)\rceil}.$$

For any $w \in \mathrm{Val}_{X,x}$, we have $a(w,X,B), a(w,X,B_k) \geq 0$ by Lemma 2.3(3), and $\mathrm{vol}(w)$ does not depend on the boundary, so (4.1) implies that

$$(1-\delta_k)^d \widehat{\mathrm{vol}}_{X,B}(w) \leq \widehat{\mathrm{vol}}_{X,B_k}(w) \leq (1+\delta_k)^d \widehat{\mathrm{vol}}_{X,B}(w),$$

where all three terms are $+\infty$ when $a(w,X,B) = +\infty$. Taking the infimum over all $w \in \mathrm{Val}_{X,x}$ and using $\widehat{\mathrm{vol}}_{X,B}(u) = \widehat{\mathrm{vol}}(X \ni x, B)$, we obtain

$$\widehat{\mathrm{vol}}(X \ni x, B_k) \geq (1-\delta_k)^d \widehat{\mathrm{vol}}(X \ni x, B) = (1-\delta_k)^d \widehat{\mathrm{vol}}_{X,B}(u) \geq \left(\frac{1-\delta_k}{1+\delta_k}\right)^d \widehat{\mathrm{vol}}_{X,B_k}(u). \tag{4.2}$$

Since $(X \ni x, B_k)$ is klt, $\widehat{\mathrm{vol}}(X \ni x, B_k) > 0$ (see the sentence after [XZ24, Definition 2.1]). Together with the inequalities between normalized volumes above for $w = u$, we obtain

$$0 < \widehat{\mathrm{vol}}(X \ni x, B_k) \leq \widehat{\mathrm{vol}}_{X,B_k}(u) \leq (1+\delta_k)^d \widehat{\mathrm{vol}}_{X,B}(u) < +\infty,$$

so the quotient $\widehat{\mathrm{vol}}(X \ni x, B_k)/\widehat{\mathrm{vol}}_{X,B_k}(u)$ used in **Step 3** is a positive real number.

Step 3. In this step we show that for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D' \geq 0$ on X, any real number $t' \geq 0$ such that $t' \cdot u(D') \leq c_0(d) a(u,X,B)$, and any prime divisor F over X whose center on X contains x, we have

$$a(F,X,B) \geq t' \,\mathrm{mult}_F\, g^* D', \tag{4.3}$$

where $g : X' \to X$ is a projective birational morphism from a normal variety X' such that F is a prime divisor on X'.

Let $\rho_k := \frac{(1-\delta_k)^{d+1}}{(1+\delta_k)^d}$. By (4.1),

$$(1-\delta_k) a(u,X,B) \leq a(u,X,B_k) \leq (1+\delta_k) a(u,X,B) < +\infty.$$

Together with (4.2), we have

$$\rho_k t' \cdot u(D') \leq \rho_k c_0(d) a(u,X,B) \leq c_0(d) \frac{\widehat{\mathrm{vol}}(X \ni x, B_k)}{\widehat{\mathrm{vol}}_{X,B_k}(u)} a(u,X,B_k).$$

Since (X,B_k) is klt, B_k is a $\mathbb{Q}$-divisor, and $K_X + B_k$ is $\mathbb{Q}$-Cartier, [Zhu24a, Lemma 3.4] implies that $(X \ni x, B_k + \rho_k t' D')$ is lc. Since the center of F on X contains x, F is a prime divisor over any open neighborhood of x. Thus, by Definition 2.1,

$$a(F,X,B_k) - \rho_k t' \,\mathrm{mult}_F\, g^* D' = a(F,X,B_k + \rho_k t' D') \geq 0.$$

As $k \to +\infty$, we have $\rho_k \to 1$, and $a(F,X,B_k) \to a(F,X,B)$ by (4.1) applied to the divisorial valuation defined by F. This implies (4.3).

Step 4. In this step we approximate D by $\mathbb{Q}$-divisors and conclude the proof.

Since D is $\mathbb{R}$-Cartier, possibly replacing X with an affine open neighborhood of x again (the conclusions of **Steps 1–3** remain valid after shrinking X), we may assume that

$$D = \sum_l e_l \,\mathrm{div}(\psi_l)$$

for finitely many real numbers e_l and rational functions ψ_l on X. Write $D = \sum_j \kappa_j Q_j$, where the Q_j are distinct prime divisors and $\kappa_j > 0$ for any j. The equation $\sum_j y_j Q_j = \sum_l z_l \,\mathrm{div}(\psi_l)$ in the unknowns (y_j, z_l) is a homogeneous system of linear equations with integer coefficients, and (κ_j, e_l) is a real solution. As in **Step 1**, there exist rational solutions $(\kappa_{k,j}, e_{k,l})$, $k = 1, 2, \ldots,$

such that $\kappa_{k,j} > 0$ for any k and j, and $(\kappa_{k,j}, e_{k,l}) \to (\kappa_j, e_l)$ as $k \to +\infty$. Let $D_k := \sum_j \kappa_{k,j} Q_j$. Then $D_k = \sum_l e_{k,l} \operatorname{div}(\psi_l) \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. By Definition-Lemma 2.2,

$$u(D_k) = \sum_l e_{k,l} u(\psi_l) \to \sum_l e_l u(\psi_l) = u(D),$$

and for any prime divisor F over X and g as in **Step 3**,

$$\operatorname{mult}_F g^* D_k = \sum_l e_{k,l} \operatorname{mult}_F g^* \operatorname{div}(\psi_l) \to \operatorname{mult}_F g^* D$$

as $k \to +\infty$.

Let $t \geq 0$ be a real number such that $t \cdot u(D) \leq c_0 \cdot a(u, X, B)$, let F be a prime divisor over X whose center on X contains x, and let $\theta \in [0, 1)$ be a real number. Since $a(u, X, B) > 0$ and $c_0 \leq c_0(d)$, we have $\theta t \cdot u(D) < c_0(d) a(u, X, B)$. Thus $\theta t \cdot u(D_k) \leq c_0(d) a(u, X, B)$ for any $k \gg 0$, and **Step 3** implies that $a(F, X, B) \geq \theta t \operatorname{mult}_F g^* D_k$ for any $k \gg 0$. Letting $k \to +\infty$ and then $\theta \to 1$, we obtain

$$a(F, X, B + tD) = a(F, X, B) - t \operatorname{mult}_F g^* D \geq 0. \tag{4.4}$$

Let $g : X' \to X$ be a log resolution of $(X, B + tD)$, and write $K_{X'} + \Delta' := g^*(K_X + B + tD)$. Let V be the complement in X of the union of the images of the prime divisors F on X' such that $\operatorname{mult}_F \Delta' > 1$. By (4.4), the image of any such F does not contain x, so V is an open neighborhood of x, and all coefficients of Δ' over V are at most 1. Thus $(V, (B + tD)|_V)$ is lc [KM98, Lemma 2.30 and Corollary 2.31(3)], that is, $(X \ni x, B + tD)$ is lc. Here [KM98, Lemma 2.30 and Corollary 2.31(3)] are formulated for $\mathbb{Q}$-divisors, and their proofs apply to $\mathbb{R}$-divisors without change. □

Remark 4.2. The assumption that u minimizes the normalized volume cannot be removed from Lemma 4.1. For coordinates z_1, z_2 of $\mathbb{A}^2$, the prime divisor $D := \{z_1 = 0\}$, and the monomial valuation u with weights 1 and m on z_1 and z_2 for a positive integer m, we have $a(u, \mathbb{A}^2, 0) = 1 + m$ and $u(D) = 1$ [Li18, Example 5.1], so without the assumption, the conclusion would imply that $(\mathbb{A}^2 \ni 0, c_0(1 + m)D)$ is lc, which forces $c_0 \leq \frac{1}{1+m}$ for any positive integer m (cf. the discussion after [Zhu24a, Corollary 3.5]). On the other hand, for a hyperplane $H \subset \mathbb{A}^d$ through 0 and the divisorial valuation u defined by the exceptional divisor of the blow-up of $\mathbb{A}^d$ at 0, which minimizes the normalized volume [Li18, Proposition 5.2], we have $u(H) = 1$ and $a(u, \mathbb{A}^d, 0) = d$, so Lemma 4.1 implies that $(\mathbb{A}^d \ni 0, c_0 dH)$ is lc, hence $c_0 \leq \frac{1}{d}$.

Proposition 4.3. *Let d be a positive integer and $c_0 \in (0, 1]$ the real number given by Lemma 4.1. Let $(X \ni x, B; \xi)$ be a d-dimensional K-semistable polarized log Fano cone singularity, T the acting torus, and $X = \operatorname{Spec} R$. Let $K_X \geq 0$ be a T-invariant canonical divisor of X, $\bar{B} \geq B$ a T-invariant $\mathbb{Q}$-divisor on X, and N a positive integer such that $N\bar{B}$ is a Weil divisor. Let $\lambda > \frac{1}{c_0}$ be a real number and $0 \neq s \in H^0(X, \mathcal{O}_X(-N(K_X + \bar{B})))$ an element such that $(X \ni x, B + \lambda G)$ is lc, where $G := \frac{1}{N} \operatorname{div}(s) - (K_X + B)$. Then there exists $\alpha \in M$ with $s_\alpha \neq 0$ and $\langle \xi, \alpha \rangle = \operatorname{wt}_\xi(s)$ such that $B^+ := \frac{1}{N} \operatorname{div}(s_\alpha) - K_X$ satisfies the following (see Definition 2.4 for M, s_α, and wt_ξ).*

1. *B^+ is a T-invariant $\mathbb{Q}$-divisor, NB^+ is a Weil divisor, $B^+ \geq \bar{B}$, and $N(K_X + B^+)$ is a principal divisor.*
2. *(X, B^+) is klt.*
3. *For any $w \in \operatorname{Val}_{X,x}$, we have*

$$a(w, X, B^+) \geq \left(1 - \frac{1}{c_0 \lambda}\right) a(w, X, B).$$

In particular,

$$\widehat{\operatorname{vol}}(X \ni x, B^+) \geq \left(1 - \frac{1}{c_0 \lambda}\right)^d \widehat{\operatorname{vol}}(X \ni x, B).$$

Proof. By Lemma 2.6(2), (X, B) is a klt pair, so $K_X + B$ is $\mathbb{R}$-Cartier. Since $N(K_X + \bar{B}) \geq 0$ is a Weil divisor and $s \in H^0(X, \mathcal{O}_X(-N(K_X + \bar{B})))$, we have $\operatorname{div}(s) \geq N(K_X + \bar{B})$. Thus

$$G = (\bar{B} - B) + \frac{1}{N}\left(\operatorname{div}(s) - N(K_X + \bar{B})\right) \geq 0,$$

and G is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor because $\operatorname{div}(s)$ is Cartier and $K_X + B$ is $\mathbb{R}$-Cartier.

Since K_X and $\bar{B}$ are T-invariant, by Lemma 2.6(3), $s_\beta \in H^0(X, \mathcal{O}_X(-N(K_X + \bar{B})))$ for any $\beta \in M$. Since $s \neq 0$, only finitely many s_β are non-zero and at least one of them is non-zero. We fix $\alpha \in M$ such that $s_\alpha \neq 0$ and

$$\langle \xi, \alpha \rangle = \min\{\langle \xi, \beta \rangle \mid \beta \in M,\ s_\beta \neq 0\}.$$

The element α need not be unique. By the definition of wt_ξ (Definition 2.4),

$$\operatorname{wt}_\xi(s_\alpha) = \langle \xi, \alpha \rangle = \operatorname{wt}_\xi(s).$$

Let $B^+ := \frac{1}{N}\operatorname{div}(s_\alpha) - K_X$.

(1) Since $s_\alpha \in H^0(X, \mathcal{O}_X(-N(K_X + \bar{B})))$, we have $\operatorname{div}(s_\alpha) \geq N(K_X + \bar{B})$, hence

$$B^+ - \bar{B} = \frac{1}{N}\left(\operatorname{div}(s_\alpha) - N(K_X + \bar{B})\right) \geq 0.$$

Moreover, B^+ is a $\mathbb{Q}$-divisor, and

$$NB^+ = \operatorname{div}(s_\alpha) - NK_X, \qquad N(K_X + B^+) = \operatorname{div}(s_\alpha),$$

so NB^+ is a Weil divisor and $N(K_X + B^+)$ is a principal divisor. Since $s_\alpha \in R_\alpha$, s_α is a T-semi-invariant function, that is, the translate of s_α by any element of T is a non-zero constant multiple of s_α. Thus $\operatorname{div}(s_\alpha)$ is T-invariant. Since K_X is T-invariant, B^+ is T-invariant. This implies (1).

Let $G^+ := B^+ - B$. Then $G^+ = (B^+ - \bar{B}) + (\bar{B} - B) \geq 0$, and $G^+ = \frac{1}{N}\operatorname{div}(s_\alpha) - (K_X + B)$ is $\mathbb{R}$-Cartier. By Definition-Lemma 2.2,

$$\operatorname{wt}_\xi(G^+) = \frac{1}{N}\operatorname{wt}_\xi(s_\alpha) - \operatorname{wt}_\xi(K_X + B) = \frac{1}{N}\operatorname{wt}_\xi(s) - \operatorname{wt}_\xi(K_X + B) = \operatorname{wt}_\xi(G).$$

Since $(X \ni x, B + \lambda G)$ is lc and $\lambda G \geq 0$ is $\mathbb{R}$-Cartier, by Lemma 2.3(1)(3),

$$0 \leq a(\operatorname{wt}_\xi, X, B + \lambda G) = a(\operatorname{wt}_\xi, X, B) - \lambda \operatorname{wt}_\xi(G).$$

Thus

$$c_0\lambda \cdot \operatorname{wt}_\xi(G^+) = c_0\lambda \cdot \operatorname{wt}_\xi(G) \leq c_0 \cdot a(\operatorname{wt}_\xi, X, B).$$

Since $(X \ni x, B; \xi)$ is K-semistable, $\widehat{\operatorname{vol}}_{X,B}(\operatorname{wt}_\xi) = \widehat{\operatorname{vol}}(X \ni x, B)$. Since $(X \ni x, B)$ is a klt pair of dimension d and $G^+ \geq 0$ is $\mathbb{R}$-Cartier, Lemma 4.1 with $u = \operatorname{wt}_\xi$, $D = G^+$, and $t = c_0\lambda$ implies that $(X \ni x, B + c_0\lambda G^+)$ is lc. Let $\mu := \frac{1}{c_0\lambda}$. Since $\lambda > \frac{1}{c_0}$, we have $0 < \mu < 1$, and

$$B^+ = B + G^+ = (1 - \mu)B + \mu(B + c_0\lambda G^+).$$

(2) Let U be an open neighborhood of x such that $(U, B|_U)$ is klt and $(U, (B + c_0\lambda G^+)|_U)$ is lc. For any prime divisor E over U, since $K_X + B^+ = (1 - \mu)(K_X + B) + \mu(K_X + B + c_0\lambda G^+)$ and pulling back $\mathbb{R}$-Cartier $\mathbb{R}$-divisors is linear, Definition 2.1 implies that

$$a(E, X, B^+) = (1 - \mu)a(E, X, B) + \mu a(E, X, B + c_0\lambda G^+) > 0,$$

because $a(E, X, B) > 0$, $a(E, X, B + c_0\lambda G^+) \geq 0$, and $1 - \mu > 0$. Thus $(X \ni x, B^+)$ is klt. Since $B^+ \geq 0$ is T-invariant, by Lemma 2.6(2), (X, B^+) is klt. This implies (2).

(3) Let $w \in \operatorname{Val}_{X,x}$. By Lemma 2.3(2), applied to $(X \ni x, B)$, $(X \ni x, B + c_0\lambda G^+)$, and $t = 1 - \mu \in (0, 1)$, and by Lemma 2.3(3), applied to the lc pair $(X \ni x, B + c_0\lambda G^+)$, we have

$$a(w, X, B^+) = (1 - \mu)a(w, X, B) + \mu a(w, X, B + c_0\lambda G^+) \geq (1 - \mu)a(w, X, B).$$

This implies the first inequality of (3).

By (2), $(X \ni x, B^+)$ is klt. For any $w \in \mathrm{Val}_{X,x}$ with $a(w,X,B) < +\infty$, we have $0 \le a(w,X,B^+) \le a(w,X,B) < +\infty$ by Lemma 2.3(1)(3), and the first inequality of (3) implies that

$$\widehat{\mathrm{vol}}_{X,B^+}(w) = a(w,X,B^+)^d \,\mathrm{vol}(w) \ge (1-\mu)^d a(w,X,B)^d \,\mathrm{vol}(w) = (1-\mu)^d \widehat{\mathrm{vol}}_{X,B}(w),$$

and for any $w \in \mathrm{Val}_{X,x}$ with $a(w,X,B) = +\infty$, both $\widehat{\mathrm{vol}}_{X,B^+}(w)$ and $\widehat{\mathrm{vol}}_{X,B}(w)$ are $+\infty$. Taking the infimum over all $w \in \mathrm{Val}_{X,x}$, we obtain

$$\widehat{\mathrm{vol}}(X \ni x, B^+) \ge (1-\mu)^d \widehat{\mathrm{vol}}(X \ni x, B) = \left(1 - \frac{1}{c_0 \lambda}\right)^d \widehat{\mathrm{vol}}(X \ni x, B).$$

□

5. A boundedness criterion

In this section, we prove the boundedness criterion Proposition 5.1, which reduces Theorem 1.2 to constructing B^+ as in the sketch of the proof in Section 1.

Proposition 5.1. *Let d, N be two positive integers and ϵ_0 a positive real number. Let $\mathcal{A}$ be a set of d-dimensional K-semistable polarized log Fano cone singularities $(X \ni x, B; \xi)$. Assume that for any $(X \ni x, B; \xi) \in \mathcal{A}$ with acting torus T, there exists a T-invariant $\mathbb{Q}$-divisor $B^+ \ge B$ on X such that NB^+ is a Weil divisor, (X, B^+) is klt, and $\widehat{\mathrm{vol}}(X \ni x, B^+) \ge \epsilon_0$. Then the set of triples $(X, \operatorname{Supp} B; x)$, where $(X \ni x, B; \xi) \in \mathcal{A}$, is log bounded.*

Remark 5.2. Proposition 5.1 does not ask $(X \ni x, B^+; \xi)$ to be K-semistable. This is necessary: for the $\mathbb{Q}$-divisor B^+ provided by Proposition 4.3, there may exist no Reeb vector ξ with respect to the acting torus such that $(X \ni x, B^+; \xi)$ is K-semistable.

For example, let $X = \mathbb{A}^2$ with coordinates z_1, z_2, and let $T = \mathbb{G}_m$ act on X by $t \cdot (z_1, z_2) = (tz_1, tz_2)$. The Reeb vectors on X are the positive real numbers ξ, and $\mathrm{wt}_\xi = \xi \,\mathrm{wt}_1$. In Proposition 4.3, take $B = 0$, $K_X = 0$, $\bar{B} = 0$, a real number $\lambda > \frac{1}{c_0}$, an integer $N \ge \lambda$, and $s := z_1 + z_2^2$. Then $\lambda G = \frac{\lambda}{N} C$, where $C := \{z_1 + z_2^2 = 0\}$ is a smooth curve and $\frac{\lambda}{N} \le 1$, so $(X \ni 0, \lambda G)$ is lc and the hypotheses of Proposition 4.3 hold (and $(X \ni 0, 0; \xi)$ is K-semistable for any Reeb vector ξ by [Li18, Proposition 5.2], since wt_ξ is a positive multiple of wt_1, the valuation of the exceptional divisor of the blow-up of X at 0, and the normalized volume is invariant under rescaling). The components of s are z_1 of weight 1 and z_2^2 of weight 2, so $\mathrm{wt}_\xi(s) = \xi$ for any Reeb vector ξ, and the element α provided by Proposition 4.3 satisfies $s_\alpha = z_1$. Thus $B^+ = cP$ with $P := \{z_1 = 0\}$ and $c := \frac{1}{N}$, and $c \in (0,1)$ since $N \ge \lambda > \frac{1}{c_0} \ge 1$. Since $a(\mathrm{wt}_1, X, 0) = 2$, $\mathrm{wt}_1(P) = 1$, and $\mathrm{vol}(\mathrm{wt}_1) = 1$ [Li18, Example 5.1], Lemma 2.3(1) implies that

$$\widehat{\mathrm{vol}}_{X,B^+}(\mathrm{wt}_\xi) = \widehat{\mathrm{vol}}_{X,B^+}(\mathrm{wt}_1) = (2-c)^2$$

for any Reeb vector ξ. Let v be the monomial valuation in the coordinates z_1, z_2 with $v(z_1) = 1$ and $v(z_2) = 1 - c$. By [Li18, Example 5.1], $a(v, X, 0) = 2 - c$ and $\mathrm{vol}(v) = \frac{1}{1-c}$, so $a(v, X, B^+) = 2 - 2c$ and $\widehat{\mathrm{vol}}_{X,B^+}(v) = 4(1-c)$. Since

$$4(1-c) < (2-c)^2,$$

we have $\widehat{\mathrm{vol}}(X \ni 0, B^+) < \widehat{\mathrm{vol}}_{X,B^+}(\mathrm{wt}_\xi)$, and $(X \ni 0, B^+; \xi)$ is not K-semistable for any Reeb vector ξ.

Therefore Proposition 5.1 only asks for a lower bound of $\widehat{\mathrm{vol}}(X \ni x, B^+)$, which suffices for [XZ24, Theorem 1.4] through the volume ratio $\Theta(X, B^+; \xi)$.

In the proof of Proposition 5.1, [XZ24, Theorem 1.4] implies the log boundedness of triples $(X, D_X; x)$, where $D_X \subset X$ is a closed subset with $\operatorname{Supp} B \subset \operatorname{Supp} B^+ \subset D_X$, and D_X may have more irreducible components than $\operatorname{Supp} B$. The following lemma shows that forgetting some irreducible components, while keeping the marked point, preserves log boundedness, provided that the fibers of the family in Definition 2.7 over all points of the base are normal and the intersection of the closed subset with any fiber is the support of a divisor on that fiber.

Lemma 5.3. *Let $\mathcal{C}$ be a set of triples $(X, D; x)$, where X is a variety, $D \subset X$ is a closed subset, and $x \in X$ is a closed point. Assume that there exist a morphism $f : \mathcal{X} \to S$, a closed subset $\mathcal{D} \subset \mathcal{X}$, and a section σ of f as in Definition 2.7 for $\mathcal{C}$, such that for any point $t \in S$ (not necessarily closed), the fiber $f^{-1}(t)$ is normal and $\mathcal{D} \cap f^{-1}(t)$ is the support of a divisor on $f^{-1}(t)$. Let $\mathcal{C}'$ be the set of triples $(X, D'; x)$, where $(X, D; x) \in \mathcal{C}$ and D' is a (possibly empty) union of irreducible components of D. Then $\mathcal{C}'$ is log bounded.*

Proof. Write $\mathcal{X}_t := f^{-1}(t)$ for any $t \in S$. For any $(X, D; x) \in \mathcal{C}$ with closed point $t \in S$ and isomorphism $g : X \to \mathcal{X}_t$ as in Definition 2.7, g maps any union of irreducible components of D onto a union of irreducible components of $\mathcal{D} \cap \mathcal{X}_t$. The restrictions of f, $\mathcal{D}$, and σ over any reduced closed subscheme of S satisfy the assumptions of the lemma, since their fibers are fibers of f. Moreover, the disjoint union of finitely many schemes of finite type over $\mathbb{C}$ is of finite type over $\mathbb{C}$, and closed subsets and sections can be taken on each piece. Thus, by Noetherian induction on S, it suffices to show the following: if $S \neq \emptyset$, then there exist a non-empty open subset $S_0 \subset S$ and finitely many morphisms $f_J : \mathcal{X}_J \to S_J$ with closed subsets $\mathcal{D}_J \subset \mathcal{X}_J$ and sections σ_J as in Definition 2.7, such that for any closed point $t \in S_0$ and any union D' of irreducible components of $\mathcal{D} \cap \mathcal{X}_t$, there exist J, a closed point $t' \in S_J$, and an isomorphism $g' : \mathcal{X}_t \to (\mathcal{X}_J)_{t'}$ such that $g'(D') = \mathcal{D}_J \cap (\mathcal{X}_J)_{t'}$ and $g'(\sigma(t)) = \sigma_J(t')$.

Let S_0 be a non-empty open subset of S that intersects only one irreducible component of S, and equip S_0 with the reduced structure. Then S_0 is integral. Possibly shrinking S_0, there exist an integral scheme S' and a finite surjective morphism $h : S' \to S_0$ such that, for the base change $f' : \mathcal{X}' := \mathcal{X} \times_{S_0} S' \to S'$ with the pullback σ' of σ and the inverse image $\mathcal{D}' \subset \mathcal{X}'$ of $\mathcal{D}$, any irreducible component of the generic fiber of $\mathcal{D}' \to S'$ is geometrically irreducible. We take for h the normalization of S_0 in a finite extension of the function field of S_0 over which the irreducible components of the geometric generic fiber of $\mathcal{D} \to S_0$ are defined [Gro65, Corollaire 4.6.8], and h is finite by [Har77, Theorem I.3.9A]. Here and below, shrinking S_0 means replacing S_0 with a non-empty open subset U and S' with $h^{-1}(U)$. For any non-empty open subset $V \subset S'$, we have $\dim h(S' \backslash V) < \dim S' = \dim S_0$, so $h^{-1}(U) \subset V$ for $U := S_0 \backslash h(S' \backslash V)$. Possibly shrinking S_0 again, we may assume that S_0 is affine and S' is non-singular. Since the characteristic is zero, by generic smoothness [Har77, Corollary III.10.7], possibly shrinking S_0 again, h is smooth, hence étale since h is finite.

Since h is étale, for any $t' \in S'$ with $t := h(t')$, the morphism $\mathcal{X}'_{t'} \to \mathcal{X}_t$ is étale. Thus $\mathcal{X}'_{t'}$ is normal, and $\mathcal{D}' \cap \mathcal{X}'_{t'}$, the inverse image of $\mathcal{D} \cap \mathcal{X}_t$, is the support of the pullback of a divisor on $\mathcal{X}_t$. Let η' be the generic point of S', and let $\mathcal{D}_1, \dots, \mathcal{D}_r$ be the closures in $\mathcal{X}'$ of the irreducible components of $\mathcal{D}'_{\eta'} := \mathcal{D}' \cap \mathcal{X}'_{\eta'}$, with the reduced structures. Then $\mathcal{D}'_{\eta'} = \bigcup_{i=1}^{r} (\mathcal{D}_i)_{\eta'}$, and any $(\mathcal{D}_i)_{\eta'}$ is geometrically irreducible and reduced, hence geometrically integral since the characteristic is zero. By [Gro66, Théorème 9.7.7], the set of points $t \in S'$ such that $(\mathcal{D}_i)_t$ is geometrically integral is locally constructible, and it contains η'. Thus, possibly shrinking S_0, $(\mathcal{D}_i)_t$ is geometrically integral for any $t \in S'$ and any i.

The generic fiber $\mathcal{X}'_{\eta'}$ is normal, so the connected components of $\mathcal{X}'_{\eta'}$ are irreducible. Any $(\mathcal{D}_i)_{\eta'}$ is a prime divisor on the connected component of $\mathcal{X}'_{\eta'}$ containing $(\mathcal{D}_i)_{\eta'}$. Thus for $i \neq j$, either $(\mathcal{D}_i)_{\eta'}$ and $(\mathcal{D}_j)_{\eta'}$ are contained in different connected components, and $(\mathcal{D}_i \cap \mathcal{D}_j)_{\eta'} = \emptyset$, or they are distinct prime divisors on one connected component Z of $\mathcal{X}'_{\eta'}$, so $(\mathcal{D}_i)_{\eta'}$ and $(\mathcal{D}_j)_{\eta'}$ both have dimension $\dim Z - 1$ and $(\mathcal{D}_i \cap \mathcal{D}_j)_{\eta'}$ is a proper closed subset of $(\mathcal{D}_i)_{\eta'}$. In both cases,

$$\dim(\mathcal{D}_i \cap \mathcal{D}_j)_{\eta'} < \dim(\mathcal{D}_i)_{\eta'},$$

where $\dim \emptyset := -1$. By [Har77, Exercise II.3.22] and generic flatness [Gro65, Théorème 6.9.1], applied to $\mathcal{D}_i \to S'$ and to the irreducible components of $\mathcal{D}_i \cap \mathcal{D}_j$ dominating S', and removing the closures of the images of the other irreducible components of $\mathcal{D}_i \cap \mathcal{D}_j$, possibly shrinking S_0, we have

$$\dim(\mathcal{D}_i \cap \mathcal{D}_j)_t \leq \dim(\mathcal{D}_i \cap \mathcal{D}_j)_{\eta'} < \dim(\mathcal{D}_i)_{\eta'} = \dim(\mathcal{D}_i)_t$$

for any $t \in S'$ and any $i \neq j$. In particular, $(\mathcal{D}_i)_t \not\subset (\mathcal{D}_j)_t$ for any $i \neq j$.

The set $\mathcal{D}'\setminus\bigcup_{i=1}^r \mathcal{D}_i$ is locally closed in $\mathcal{X}'$, so the image of $\mathcal{D}'\setminus\bigcup_{i=1}^r \mathcal{D}_i$ in S' is constructible [Har77, Exercise II.3.19]. Since $\mathcal{D}'_{\eta'}=\bigcup_{i=1}^r(\mathcal{D}_i)_{\eta'}$, the image does not contain η', so the closure of the image is a proper closed subset of S'. Possibly shrinking S_0, we may assume that

$$\mathcal{D}'\cap\mathcal{X}'_t=\bigcup_{i=1}^r(\mathcal{D}_i)_t$$

for any $t\in S'$. Thus for any closed point $t'\in S'$, the irreducible components of $\mathcal{D}'\cap\mathcal{X}'_{t'}$ are exactly the pairwise distinct closed subsets $(\mathcal{D}_1)_{t'},\dots,(\mathcal{D}_r)_{t'}$. If $r=0$, then $\mathcal{D}'\cap\mathcal{X}'_{t'}=\emptyset$.

For any subset $J\subset\{1,\dots,r\}$, let $f_J:=f'$, $\mathcal{D}_J:=\bigcup_{i\in J}\mathcal{D}_i$ (so $\mathcal{D}_\emptyset=\emptyset$), and $\sigma_J:=\sigma'$. Let $t\in S_0$ be a closed point and D' a union of irreducible components of $\mathcal{D}\cap\mathcal{X}_t$. Since h is finite and surjective and $\mathbb{C}$ is algebraically closed, there exists a closed point $t'\in S'$ such that $h(t')=t$, and the projection $\mathcal{X}'_{t'}\to\mathcal{X}_t$ is an isomorphism which maps $\mathcal{D}'\cap\mathcal{X}'_{t'}$ onto $\mathcal{D}\cap\mathcal{X}_t$ and $\sigma'(t')$ to $\sigma(t)$. Thus there exists J such that the projection maps $\mathcal{D}_J\cap\mathcal{X}'_{t'}$ onto D', and the inverse of the projection is the isomorphism g' that we need. □

Proof of Proposition 5.1. Let $\Gamma_N:=\{\frac{j}{N}\mid 0\le j\le N-1\}$. Let $(X\ni x,B;\xi)\in\mathcal{A}$ with acting torus T, and let $B^+\ge B$ be a T-invariant $\mathbb{Q}$-divisor on X as in the assumption. Since (X,B^+) is klt, any coefficient of B^+ is less than 1. Since NB^+ is a Weil divisor and $B^+\ge B\ge 0$, we have $B^+\in\Gamma_N$. Since B^+ is T-invariant, (X,B^+) is klt, and ξ is a Reeb vector on X, $(X\ni x,B^+;\xi)$ is a polarized log Fano cone singularity.

By Lemma 2.6(2), (X,B) is a klt pair, so K_X+B is $\mathbb{R}$-Cartier. Since (X,B^+) is klt, K_X+B^+ is $\mathbb{R}$-Cartier. Thus $F:=B^+-B\ge 0$ is $\mathbb{R}$-Cartier. By Definition 2.5, since $(X\ni x,B;\xi)$ is K-semistable, and by [LX19, Lemma A.1] (see [Zhu24b, Lemma 2.6] for $\mathbb{R}$-divisors), respectively,

$$\widehat{\mathrm{vol}}_{X,B}(\mathrm{wt}_\xi)=\widehat{\mathrm{vol}}(X\ni x,B)\le d^d.$$

In particular, $a(\mathrm{wt}_\xi,X,B)<+\infty$. By Lemma 2.3(1)(3),

$$0\le a(\mathrm{wt}_\xi,X,B^+)=a(\mathrm{wt}_\xi,X,B)-\mathrm{wt}_\xi(F)\le a(\mathrm{wt}_\xi,X,B).$$

Taking d-th powers and multiplying by $\mathrm{vol}(\mathrm{wt}_\xi)$, we obtain

$$\widehat{\mathrm{vol}}_{X,B^+}(\mathrm{wt}_\xi)\le\widehat{\mathrm{vol}}_{X,B}(\mathrm{wt}_\xi)=\widehat{\mathrm{vol}}(X\ni x,B)\le d^d.$$

Thus

$$\Theta(X,B^+;\xi)=\frac{\widehat{\mathrm{vol}}(X\ni x,B^+)}{\widehat{\mathrm{vol}}_{X,B^+}(\mathrm{wt}_\xi)}\ge\frac{\epsilon_0}{d^d}.$$

By [XZ24, Theorem 1.4], for any positive integer n, any finite set $I\subset[0,1]$, and any two positive real numbers δ_1,δ_2, the n-dimensional polarized log Fano cone singularities whose coefficients are in I, local volume is at least δ_1, and volume ratio is at least δ_2 form a bounded set in the sense of [XZ24, Definition 2.16]. Thus, by [XZ24, Theorem 1.4] with $n=d$, $I=\Gamma_N$, $\delta_1=\epsilon_0$, and $\delta_2=\epsilon_0/d^d$, the set of the polarized log Fano cone singularities $(X\ni x,B^+;\xi)$, where $(X\ni x,B;\xi)\in\mathcal{A}$, is bounded in the sense of [XZ24, Definition 2.16]. That is, there exist finitely many $\mathbb{R}$-Gorenstein families of klt singularities $f_j:(\mathcal{X}_j,\mathcal{D}_j)\to S_j$ [XZ24, Definition 2.14] over bases S_j of finite type over $\mathbb{C}$, $1\le j\le m$, with sections σ_j and fiberwise good torus actions, such that any $(X\ni x,B^+;\xi)$ above is isomorphic to a fiber of some f_j. By the conventions of [XZ24, Section 2.1], the variety of a singularity is affine, and the marked point is contained in the support of the boundary whenever the boundary is non-zero. Both conditions hold for $(X\ni x,B^+)$: X is affine, and if $B^+\ne 0$, then $\mathrm{Supp}\,B^+$ is a non-empty closed T-invariant subset of X, so $x\in\mathrm{Supp}\,B^+$ by Lemma 2.6(1). Thus for any $(X\ni x,B;\xi)\in\mathcal{A}$, there exist j, a closed point $t\in S_j$, and an isomorphism $g:X\to(\mathcal{X}_j)_t$ of affine varieties such that $g(x)=\sigma_j(t)$ and $g_*B^+=\mathcal{D}_{j,t}$, where $\mathcal{D}_{j,t}$ is the $\mathbb{R}$-divisor on $(\mathcal{X}_j)_t$ induced by $\mathcal{D}_j$.

Let S be the disjoint union of $S_1,\dots,S_m$, $f:\mathcal{X}\to S$ the disjoint union of $f_1,\dots,f_m$, $\mathcal{D}\subset\mathcal{X}$ the disjoint union of $\mathrm{Supp}\,\mathcal{D}_1,\dots,\mathrm{Supp}\,\mathcal{D}_m$, and σ the disjoint union of $\sigma_1,\dots,\sigma_m$. Since the disjoint union of finitely many schemes of finite type over $\mathbb{C}$ is of finite type over $\mathbb{C}$, and the

closed subset and the section are taken on each piece, f, $\mathcal{D}$, and σ are as in Definition 2.7. By [XZ24, Definition 2.14], $\mathcal{X}_j$ is flat over S_j, and the fibers of f_j over the points of S_j are connected, normal, and not contained in $\operatorname{Supp}\mathcal{D}_j$. Thus any fiber of f is normal. Moreover, for any $t \in S_j$, by a dimension count [Har77, Proposition III.9.5 and Exercise II.3.22], $\operatorname{Supp}\mathcal{D}_j \cap (\mathcal{X}_j)_t$ is either empty or of pure codimension one in $(\mathcal{X}_j)_t$, so it is the support of a divisor on $(\mathcal{X}_j)_t$.

For any $(X \ni x, B; \xi) \in \mathcal{A}$, fix an index j, a closed point $t \in S_j$, and an isomorphism $g : X \to (\mathcal{X}_j)_t$ such that $g(x) = \sigma_j(t)$ and $g_*B^+ = \mathcal{D}_{j,t}$, and let $D_X := g^{-1}(\mathcal{D} \cap (\mathcal{X}_j)_t)$. Then the morphism f, the closed subset $\mathcal{D}$, and the section σ are as in Definition 2.7 for the set $\mathcal{C}$ of triples $(X, D_X; x)$, where $(X \ni x, B; \xi) \in \mathcal{A}$, and they satisfy the assumptions of Lemma 5.3. Since $0 \le B \le B^+$, $g_*B^+ = \mathcal{D}_{j,t}$, and $\operatorname{Supp}\mathcal{D}_{j,t} \subset \mathcal{D} \cap (\mathcal{X}_j)_t$, we have $\operatorname{Supp} B \subset \operatorname{Supp} B^+ \subset D_X$. Since D_X is either empty or of pure codimension one in X, any irreducible component of $\operatorname{Supp} B$ is an irreducible component of D_X. Thus $\operatorname{Supp} B$ is a (possibly empty) union of irreducible components of D_X. By Lemma 5.3, the set of triples $(X, \operatorname{Supp} B; x)$, where $(X \ni x, B; \xi) \in \mathcal{A}$, is log bounded. □

6. Proofs of the main theorems

In this section, we prove Theorem 1.2 and Corollary 1.3.

Proof of Theorem 1.2. Let $c_0 \in (0, 1]$ be the real number given by Lemma 4.1, which depends only on d, and let $\lambda := \frac{2}{c_0}$. Then $\lambda > 1$, $\lambda > \frac{1}{c_0}$, and

$$1 - \frac{1}{c_0\lambda} = \frac{1}{2}.$$

Let $\tau, \delta > 0$ and $q, N \in \mathbb{N}^+$ be the numbers given by Proposition 3.1 for $d, \epsilon, \eta, \lambda$. Since λ depends only on d, the numbers τ, δ, q, N depend only on d, ϵ, η. This is where we use that λ is arbitrary in Proposition 3.1.

Let $\mathcal{A}$ be the set of d-dimensional K-semistable polarized log Fano cone singularities $(X \ni x, B; \xi)$ such that $\widehat{\operatorname{vol}}(X \ni x, B) \ge \epsilon$ and $\operatorname{mult}_P B \ge \eta$ for any prime divisor $P \subset \operatorname{Supp} B$. By Definition 2.5, the triples in Theorem 1.2 are exactly the triples $(X, \operatorname{Supp} B; x)$, where $(X \ni x, B; \xi) \in \mathcal{A}$. Thus, by Proposition 5.1 applied to d, N, and $\epsilon_0 := 2^{-d}\epsilon$, it suffices to show that for any $(X \ni x, B; \xi) \in \mathcal{A}$ with acting torus T, there exists a T-invariant $\mathbb{Q}$-divisor $B^+ \ge B$ on X such that NB^+ is a Weil divisor, (X, B^+) is klt, and $\widehat{\operatorname{vol}}(X \ni x, B^+) \ge 2^{-d}\epsilon$.

Fix $(X \ni x, B; \xi) \in \mathcal{A}$ with acting torus T, and write $X = \operatorname{Spec} R$. By Lemma 2.6(2), (X, B) is a klt pair, and by Lemma 2.6(4), there exists a T-invariant canonical divisor $K_X \ge 0$ of X. Let

$$\bar{B} := \frac{1}{q}\lfloor q(1+\tau)B \rfloor.$$

Since T is connected, T maps each of the finitely many prime components of $\operatorname{Supp} B$ onto itself, so any divisor supported in $\operatorname{Supp} B$, in particular $\bar{B}$, is T-invariant. By Proposition 3.1(1), $\bar{B} \ge (1+\delta)B \ge B$, and $N\bar{B}$ is a Weil divisor because $q \mid N$. By Proposition 3.1(2), there exists a non-zero element $s \in H^0(X, \mathcal{O}_X(-N(K_X + \bar{B})))$ such that $(X, B + \lambda G)$ is klt near x, where $G := \frac{1}{N}\operatorname{div}(s) - (K_X + B)$. In particular, $(X \ni x, B + \lambda G)$ is lc. Since $\lambda > \frac{1}{c_0}$, by Proposition 4.3, there exists $\alpha \in M$ with $s_\alpha \ne 0$ such that $B^+ := \frac{1}{N}\operatorname{div}(s_\alpha) - K_X$ is a T-invariant $\mathbb{Q}$-divisor, NB^+ is a Weil divisor, $B^+ \ge \bar{B} \ge B$, (X, B^+) is klt, and

$$\widehat{\operatorname{vol}}(X \ni x, B^+) \ge \left(1 - \frac{1}{c_0\lambda}\right)^d \widehat{\operatorname{vol}}(X \ni x, B) = 2^{-d}\widehat{\operatorname{vol}}(X \ni x, B) \ge 2^{-d}\epsilon.$$

□

Proof of Corollary 1.3. Let d be a positive integer and ϵ, η two positive real numbers. Let $(X \ni x, B)$ be a d-dimensional K-semistable singularity in the sense of [LMS26, Definition 2.16] such that $\widehat{\operatorname{vol}}(X \ni x, B) > \epsilon$ and $\operatorname{mult}_P B > \eta$ for any prime divisor $P \subset \operatorname{Supp} B$. Let $\mathcal{K}$ be the

set of such singularities $(X \ni x, B)$. We first show that $(X \ni x, B)$ is a K-semistable log Fano cone singularity.

By [LMS26, Definitions 2.7 and 2.16], $X = \operatorname{Spec} R$ is an affine variety with a faithful action of an algebraic torus T, B is a T-invariant $\mathbb{Q}$-divisor, $(X \ni x, B)$ is klt, $R_0 = \mathbb{C}$, x is the unique T-fixed point of X, the weight cone $\omega \subset M \otimes_{\mathbb{Z}} \mathbb{R}$, generated by the weights $\alpha \in M$ with $R_\alpha \neq 0$, is full-dimensional and pointed, and there exists an element ξ of the interior of the dual cone $\omega^\vee \subset \operatorname{Hom}(M, \mathbb{R})$ such that $(X \ni x, B; \xi)$ is K-semistable in the sense of [LMS26, Definition 2.16]. Since R is generated by finitely many homogeneous elements, ω is a rational polyhedral cone. As ω is pointed, there exists $\zeta \in \operatorname{Hom}(M, \mathbb{Z})$ in the interior of $\omega^\vee$, that is, $\langle \zeta, \alpha \rangle > 0$ for any $0 \neq \alpha \in \omega$. For any integer $k \geq 0$, let $R_{(k)} := \bigoplus_{\langle \zeta, \alpha \rangle = k} R_\alpha$. Then $R = \bigoplus_{k \geq 0} R_{(k)}$ and $R_{(0)} = R_0 = \mathbb{C}$, so

$$\mathfrak{m} := \bigoplus_{k>0} R_{(k)}$$

is a maximal ideal of R. Since $\mathfrak{m}$ is homogeneous, the closed point of X defined by $\mathfrak{m}$ is T-fixed, hence is x. Let $Z \subset X$ be a non-empty closed T-invariant subset and $I_Z \subset R$ the ideal of Z. Since T is diagonalizable, I_Z is homogeneous. Since $I_Z \neq R$ and $R_0 = \mathbb{C}$, we have $I_Z \cap R_0 = 0$. Thus $I_Z \subset \bigoplus_{\alpha \neq 0} R_\alpha \subset \mathfrak{m}$, that is, $x \in Z$. In particular, x is contained in the closure of any T-orbit, and any closed point of X contained in the closure of any T-orbit is contained in the closure $\{x\}$ of the T-orbit of x. Moreover, T is non-trivial, since otherwise $R = R_0 = \mathbb{C}$, which contradicts $d \geq 1$. Since $(X \ni x, B)$ is klt, X is normal near x. Thus the non-normal locus of X is a closed T-invariant subset not containing x (the normal locus of a variety is open [Gro65, Corollaires 6.12.6 et 6.13.5]), hence is empty. Therefore, X is normal, the T-action on X is good, x is the vertex of X, and $(X \ni x, B)$ is a log Fano cone singularity.

Since ξ is contained in the interior of $\omega^\vee$, we have $\langle \xi, \alpha \rangle > 0$ for any $0 \neq \alpha \in M$ with $R_\alpha \neq 0$, that is, ξ is a Reeb vector on X. By [LMS26, Definition 2.7], the valuation in $\operatorname{Val}_{X,x}$ attached to ξ is wt_ξ, and by [LMS26, Definition 2.5], the normalized volume function and the local volume there are $\widehat{\operatorname{vol}}_{X,B}$ and $\widehat{\operatorname{vol}}(X \ni x, B)$. Since $(X \ni x, B; \xi)$ is K-semistable in the sense of [LMS26, Definition 2.16], [LMS26, Theorem 2.12] implies that wt_ξ minimizes $\widehat{\operatorname{vol}}_{X,B}$ on $\operatorname{Val}_{X,x}$, that is,

$$\widehat{\operatorname{vol}}(X \ni x, B) = \widehat{\operatorname{vol}}_{X,B}(\operatorname{wt}_\xi).$$

Hence $(X \ni x, B; \xi)$ is K-semistable in the sense of Definition 2.5, and $(X \ni x, B)$ is a K-semistable log Fano cone singularity.

Since strict inequalities imply non-strict ones, by Theorem 1.2, the set $\mathcal{T}$ of triples $(X, \operatorname{Supp} B; x)$, where $(X \ni x, B) \in \mathcal{K}$, is log bounded. Let $f : \mathcal{X} \to S$, $\mathcal{D}$, and σ be as in Definition 2.7 for $\mathcal{T}$. For any $(X \ni x, B) \in \mathcal{K}$, there exist a closed point $t \in S$ and an isomorphism $g : X \to \mathcal{X}_t$ such that $g(\operatorname{Supp} B) = \mathcal{D} \cap \mathcal{X}_t$. We denote by $\operatorname{red}(B)$ the reduced closed subscheme of X with support $\operatorname{Supp} B$. By [LMS26, Definition 2.17], taking the class of pairs (X, B) with $(X \ni x, B) \in \mathcal{K}$ and the identity as the isomorphism of germs, it suffices to show that there exist a morphism $f' : \mathcal{X}' \to S'$ of schemes of finite type over $\mathbb{C}$ and a reduced Weil divisor $\mathcal{D}' \subset \mathcal{X}'$ such that for any $(X \ni x, B) \in \mathcal{K}$, there exist a closed point $t \in S'$ and an isomorphism $X \to \mathcal{X}'_t$ mapping $\operatorname{red}(B)$ onto the scheme-theoretic fiber $\mathcal{D}'_t$. The section σ is not used. Let Σ be the set of closed points $t \in S$ for which there exist $(X \ni x, B) \in \mathcal{K}$ and an isomorphism $g : X \to \mathcal{X}_t$ with $g(\operatorname{Supp} B) = \mathcal{D} \cap \mathcal{X}_t$. Below, for $t \in \Sigma$, the letters X, B, g denote any such data.

We construct f' and $\mathcal{D}'$ by Noetherian induction on S. Let V be an irreducible component of the closure of Σ. Then $V \cap \Sigma$ is dense in V, since V is irreducible and the closure of Σ is the union of the closures of $V' \cap \Sigma$ for the finitely many irreducible components V' of the closure of Σ. It suffices to find a non-empty open subset $S^\circ \subset V$ and a reduced closed subscheme $\mathcal{D}^\circ \subset f^{-1}(S^\circ)$ of pure codimension one such that $\mathcal{D}^\circ_t = g(\operatorname{red}(B))$ for any $t \in S^\circ \cap \Sigma$. Given such S° and $\mathcal{D}^\circ$, possibly shrinking S°, we may assume that S° does not meet the other irreducible components of the closure of Σ, so S° is open in the closure of Σ, and the closure of $\Sigma \backslash S^\circ$ is a

proper closed subset of the closure of Σ. Replacing Σ with $\Sigma\backslash S^\circ$ and repeating, by Noetherian induction, finitely many such S° cover Σ, and f' and $\mathcal{D}'$ are the disjoint unions of the finitely many morphisms $f^{-1}(S^\circ) \to S^\circ$ and closed subschemes $\mathcal{D}^\circ$. Replacing S with V, with the reduced structure, Σ with $V \cap \Sigma$, and $\mathcal{X}$, $\mathcal{D}$ with their preimages, we construct S° and $\mathcal{D}^\circ$ below.

Now S is integral and $S \cap \Sigma$ is dense in S. By generic flatness [Gro65, Théorème 6.9.1], possibly shrinking S, we may assume that f is flat. Since f is flat, it is open by [Har77, Exercise III.9.1]. Let $\mathcal{W}$ be an irreducible component of $\mathcal{X}$. The points of $\mathcal{W}$ not contained in any other irreducible component of $\mathcal{X}$ form a non-empty open subset of $\mathcal{X}$, whose image is a non-empty open subset of S, so there exists a closed point $z \in \mathcal{W}$ not contained in any other irreducible component of $\mathcal{X}$ with $t := f(z) \in \Sigma$. Since $\mathcal{X}_t \cong X$ is a d-dimensional variety, by [Har77, Proposition III.9.5],

$$d = \dim_z \mathcal{X}_t = \dim_z \mathcal{X} - \dim_t S = \dim \mathcal{W} - \dim S.$$

Removing from S the closures of the images of the irreducible components of $\mathcal{D}$ not dominating S, we may assume that any irreducible component $\mathcal{E}$ of $\mathcal{D}$ dominates S. By [Har77, Exercise II.3.22(b)(e)] for $\mathcal{E} \to S$, with $\mathcal{E}$ with the reduced structure, possibly shrinking S again, we may assume that any fiber of $\mathcal{E} \to S$ is of pure dimension $\dim \mathcal{E} - \dim S$. By [Gro65, Théorème 6.9.1] for $\mathcal{E}$ with the reduced structure, possibly shrinking S again, we may assume that any $\mathcal{E} \to S$ is flat, so $f(\mathcal{E})$ is a non-empty open subset of S by [Har77, Exercise III.9.1], and $f(\mathcal{E}) \cap \Sigma \neq \emptyset$. Since $\mathcal{D} \cap \mathcal{X}_t = g(\operatorname{Supp} B)$ is empty or of pure dimension $d-1$ for any $t \in S \cap \Sigma$, $\dim \mathcal{E} \leq \dim S + d - 1$, and any irreducible component of $g(\operatorname{Supp} B)$ is contained in a fiber of some $\mathcal{E}$ with $\dim \mathcal{E} = \dim S + d - 1$. Let $\mathcal{D}^\circ$ be the union of the irreducible components $\mathcal{E}$ of $\mathcal{D}$ with $\dim \mathcal{E} = \dim S + d - 1$, with the reduced structure. Then $\mathcal{D}^\circ$ is of pure codimension one in $\mathcal{X}$, and $\mathcal{D}^\circ \cap \mathcal{X}_t = g(\operatorname{Supp} B)$ for any $t \in S \cap \Sigma$. Since $\mathcal{D}^\circ$ is reduced and we work in characteristic zero, the generic fiber of $\mathcal{D}^\circ \to S$ is geometrically reduced. By [Gro66, Théorème 9.7.7], the set of points of S over which the fiber of $\mathcal{D}^\circ$ is geometrically reduced is constructible, so it contains a non-empty open subset $S^\circ \subset S$. Thus $\mathcal{D}^\circ_t = g(\operatorname{red}(B))$ for any $t \in S^\circ \cap \Sigma$.

Therefore, the class of pairs (X, B) with $(X \ni x, B) \in \mathcal{K}$ is log bounded in the sense of [LMS26, Definition 2.17], and so is the class of singularities in Conjecture 1.1. □

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